CONSULTING SERVICES FOR PROJECT MANAGEMENT ASSISTANCE
ON
WATER RESOURCES MANAGEMENT SECTOR PROJECT

MINISTRY OF AGRICULTURE AND WATER RESOURCES
REPUBLIC OF UZBEKISTAN

OPERATION AGREEMENT
ON
THE CONSULTANCING SERVICES FOR PROJECT MANAGEMENT ASSISTANCE

BETWEEEN

DONGSHIN ENGINEERING & CONSULTANTS CO., LTD-KOREA

AND

KOREA RURAL COMMUNITY CORPORATION-KOREA

21TH APRIL 2010

ASSOCIATED CONSULTANCY AGREEMENT FOR CONSULTING SERVICES

This Agreement (hereinafter together with the appendices attached called the Agreement) is made on the 21th of April 2010 between Dongshin Engineering & Consultants Co., Ltd. of 181-1 Jamsil-dong, Songpa-gu, Seoul, Korea on the one part (hereinafter called the Consultant) and Korea Rural Community Corporation (hereinafter called the Associated Consultant) on the other part.

WHEREAS

(A) by the Agreement dated 2 May 2009 between the Republic of Uzbekistan and Asian Development Bank (hereinafter called the ADB), the ADB has agreed to make a loan to the Client for the purpose of financing Water Resources Management Sector Project (hereinafter called the Project);

(B) the Client has requested Dongshin Engineering & Consultants Co., Ltd. to provide consulting services (hereinafter called the Services) necessary for the effective implementation of the Project.

(C) the Consultant proposes to obtain support of the Associated Consultant, Engineering Consultants (PVT) LTD. (thereafter referred to as Associated Consultant) to facilitate project implementation and to this effect the Associated Consultant has agreed to provide the Services of the following specialists, on the terms and conditions set forth in this Agreement, and in accordance to the General Conditions of Contract applicable to the Services.

Specialist	**Person Month Input per Year**					
	Year 1	**Year 2**	**Year 3**	**Year 4**	**Year 5**	**Total**
Electrical Engineer/Kim Jeong-Seog	2.0	2.0	2.0			6.0
Total KRC Input (person months)						**6.0**

NOW THEREFORE the parties hereto hereby agree as follows:

ARTICLE I Services

1.01 **The *Services***

The Associated Consultant shall perform the Services under this Agreement as a contribution to the total assignment as described in the Background Information and Terms of Reference issued on the Consultant by the Client, as set forth in Appendix A hereto. The specific tasks which are anticipated will be undertaken directly or shared by the Associated Consultant's personnel are shown in Appendix B together with the Task Allocation Table for the position

1.02 **Commencement *Date***

The Consultant shall provide fifteen (15) calendar days advance notice to proceed with mobilization of each of the proposed staff. Presently anticipated mobilization dates are indicated in the Staffing Schedule (Appendix D), but will be updated from time to time by the Team Leader.

ARTICLE II Personnel

2.01 ***Personnel***

(a) The Services shall be carried out by the personnel specified in Appendix B hereof (hereinafter called the Personnel) for the respective periods of time indicated therein and in the Staffing Schedule at Appendix D. The Associated Consultant

may, with the prior approval of the Consultant and the Client, make minor adjustments in such periods as may be appropriate to ensure the efficient performance of the Services, provided that such adjustments shall not cause payments made under the Agreement to exceed the cost estimates referred to in Section 3.01.

(b) Except as the Consultant may otherwise agree, no changes shall be made in the Personnel. If for any reason beyond the reasonable control of the Associated Consultant it becomes necessary to replace any of the Personnel, the Associated Consultant shall forthwith provide as a replacement, a person of equivalent or better qualifications and to the satisfaction of the Client.

(c) In the event that any Personnel specified in Appendix B is found by the Consultant and the Client to be incompetent in discharging his assigned duties, the Consultant may request the Associated Consultant to forthwith provide as a replacement a person with acceptable qualifications and experience.

(d) Any person provided as a replacement under Sections 2.01 (b) or (c), the rate of remuneration applicable to him, and any out-of-pocket expenses incurred by reason of such replacement for which the Associated Consultant may seek reimbursement shall be subject to the prior approval of the Consultant. Except as the Consultant may otherwise agree the Associated Consultant shall bear all additional travel and other costs arising out of or incidental to any replacement. The remuneration to be paid for such person shall not exceed the remuneration which would have been payable to the person replaced.

(e) All the replacement of the Personnel shall be approved by the Client, and minimum one week (7 days) transfer period of his duties shall be required. The remuneration for this period shall be borne by the Associated Consultant.

ARTICLE III Payments to the Associated Consultant

3.01 ***Cost Estimates: Unit Rates***

(a) An estimate of the unit costs of the Services to be provided, plus out of pocket expenses payable is set forth in Appendix C.

(b) Except as may be otherwise agreed under Section 6.07 and notwithstanding any other provision of this Agreement, all payments made under this Agreement shall not exceed the unit rates in US Dollars indicated. The actual number of person months will be verified during the course of the project, and the final fee for services provided will require approval of the consultants Team Leader before payment.

3.02 ***Currency of Payment***

(a) Payment in respect of all local currency expenditures incurred by the Associated Consultant under Section 3.03 will be made in US Dollars.

3.03 ***Payments to Associated Consultant***

(a) The Consultant shall pay or reimburse to the Associated Consultant in respect of the Services, but subject to the unit rates specified in Section 3.01 (b), remuneration as set forth in Section 3.03 (b) and out-of-pocket expenses as set forth in Section 3.03 (c).

(b) Remuneration shall be determined on the basis of time actually spent by the Personnel in the performance of the Services after the effective date thereof at the rates specified in respect of such personnel in Appendix C, subject to the following.

(1) The remuneration shall include salary and related costs, costs in respect of insurance, sickness, holidays and vacation pay, pension fund and similar payments, overhead charges, cost of backstopping by home office staff and

all other expenses not specified in this Section, as well as the Associated Consultant's fee.

(2) Remuneration for periods of less than one month shall be calculated on a calendar-day basis for time spent on the project (1 day being equivalent to 1/30th of a month).

(3) Notwithstanding anything stated above, remuneration will be paid to the Associated Consultant in respect of the Personnel assigned on the Project for all national holidays in Uzbekistan.

(4) The Consultant shall not be responsible for payment of annual leave or sick leave.

(c) Out-of-pocket expenses, as shown in Appendix C, shall consist of the following types of expenses reasonably incurred by the Associated Consultant in the performance of the Services.

(1) A per diem allowance for specialists for every day in which such Personnel is required to be resident in Uzbekistan for the purpose of the Services at the agreed rates as in Appendix C per day.

(2) Costs of local transportation required for the purpose of the Services (where not provided by the Consultant or the Client), and for mobilization/demobilization, all to be substantiated by receipts.

(3) Cost of items not covered in the foregoing but which may be required by the Associated Consultant for completion of the Services, subject to the prior authorization in writing of the Client and substantiated by receipts.

3.04 ***Mode of Billing and Payment***

Billings and payments in respect of the Services shall be made as follows:

(a) As soon as practicable and not later than fifteen days after the end of each Calendar month during the period of the Services, the Associated Consultant shall submit to the Consultant duplicate, itemized statements, accompanied by receipted invoices, vouchers and other appropriate supporting materials, of the amounts payable pursuant to Section 3.03 for such month. Each such monthly statement shall distinguish that portion of the total eligible costs which pertains to remuneration from that portion which pertains to out-of-pocket expenses.

(b) The Consultant shall cause payment of the Associated Consultant's monthly statements within 30 days after receipt of monthly statements. Only such portion of a monthly statement that is not satisfactorily supported may be withheld from payment. Should any discrepancy be found to exist at any time between actual payments and costs authorized to be incurred by the Associated Consultant, the Consultant may at any time, add or subtract the difference from any subsequent payments.

(c) All payments under this Agreement shall be made to the account of the Associated Consultant as follows:

-Name of Bank : National Agricultural Cooperative Federation(NACF)
-Address of Bank : 487 Poildong, Uiwang-si, Gyeonggi-do,
National Agricultural Cooperative Federation,
KRCS-Sub Branch
-Name of Beneficiary: Korea Rural Community Corporation
-Address of Beneficiary: 98 Anyang Pangyo-ro(487 Poildong), Uiwang-si,
Gyeonggi-do, Republic of Korea,
Korea Rural Community Corporation
-Beneficiary's Account No. : 573-01-000430
-Swift Code : NACFKRSEXXX
-Tele No.: +82 31 422 4256
-Fax No.: + 82 31 422 3591

Correspond ing Bank: -
Corresponding Bank Account No.: -
Correcponding Bank SWIFT Code: -

3.05 ***Additional Work***

If, in the opinion of the Consultant or the Client, it is necessary to carry out any work outside of the Terms of Reference for the purposes of the Project in additional to the Services, such additional work may be carried out with the prior concurrence of the Client. The Associated Consultant, with the prior authorization of the Consultant, shall carry out such additional work and charge for the Personnel required on the same basis as for the Services.

ARTICLE IV Undertaking of the Consultant

4.01 ***Services, Facilities and Equipment***

The Consultant shall make available to the Associated Consultant and his Personnel, for the purposes of the Services, and free of any charge, such services, facilities and property provided through the Client. This will include but not necessarily limited to furnished office facilities, site transport, copies of all necessary project documentations and access to communication facilities.

ARTICLE V Undertakings of the Associated Consultant

5.01 ***General Standard of Performance by the Associated Consultant***

(a) The Associated Consultant's staff shall carry out their tasks with due diligence and efficiency, and shall exercise such skill and care in the performance of the Services as is consistent with recognized professional standards.

(b) The Associated Consultant shall act at all times so as to protect the interests of the Consultant and will take all reasonable steps to keep all expenses to minimum, consistent with sound and efficient practices.

5.02 ***Records***

(a) The Associated Consultant shall keep accurate and systematic records and accounts in respect of the Services in such form and detail as is customary in the profession and as shall be sufficient to establish accurately that the costs and expenditures have duly incurred.

(b) The Associated Consultant shall permit the duly authorized representative of the Consultant and, if so requested, the duly authorized representative of the Client, from time to time to inspect its records and accounts relating to the Services and to make copies thereof and shall permit the Consultant, or any person authorized by the Client, from time to time, to audit such records and accounts during and after the Services.

5.03 ***Information***

The Associated Consultant shall furnish the Consultant such information relating to the Services and the Project as the Consultant may from time to time reasonably request.

5.04 ***Assignments***

(a) Except with the prior written approval of the Consultant, the Associated Consultant shall not assign or transfer this Agreement or any part thereof nor engage any independent sub-consultant to perform any part of the Services.

(b) The approval by the Consultant to the assignment of any part of the Agreement or

to the engagement by the Associated Consultant of independent sub-consultants to perform any part of the Services shall not relieve the Associated Consultant of any of its obligations under the Agreement.

(c) In the event that any such independent sub-contractor is found by the Consultant to be incompetent in discharging his assigned duties, the Consultant may request the Associated Consultant forthwith either to provide as a replacement, a sub-consultant with qualifications and experience acceptable to the Consultant, or to resume the performance of the Services itself.

5.05 ***Confidentiality***

Except with the prior written consent of the Consultant, the Associated Consultant and its Personnel shall not at any time communicate to any person or the entity any confidential information disclosed to them for the purposes of the Services or discovered by them in the course of the Services, nor shall the Associated Consultant or the Personnel make public any information as to the recommendations formulated in the course of or as a result of the Services.

5.06 ***Prohibition on Association***

The Associated Consultant agrees that during the period of this Agreement, the Associated Consultant limits its role under the Project to the provision of the Services and hereby disqualifies itself and any other consulting engineer with which the Associated Consultant is associated or affiliated from the provision of goods or services in any capacity for the Project including bidding for any part of the Project except as the Consultant and the Client may otherwise agree.

5.07 ***Indemnifications***

(a) The Associated Consultant shall indemnify, protect and defend at the Associated Consultant's own expense, the Consultant, the Client and the employees, from and against any and all actions, claims, losses or damages arising out of any violation by the Associated Consultant or in the course of the services of any legal provisions, or any rights of third parties, in respect of literary property rights, copyrights, or patents.

(b) The Associated Consultant shall indemnify, protect and defend at the Associated Consultant's own expense, the Consultant, the Client and its employees, from and against any and all actions, claims, losses or damages arising out of Associated Consultant's failure to exercise the skill and care required under Section 5.01 (a), provided, however:

- the Associated Consultant is notified of such actions, claims, losses or damages not later than six (6) months after conclusion of the Services;
- that the ceiling on Associated Consultant's liability under this Section 5.07 (b) shall be limited to US$15,000 except that such ceiling shall not apply to actions, claims, losses or damages caused by Associated Consultant's gross negligence or reckless conduct;
- that Associated Consultant's liability under this Section 5.07 (b) shall be limited to actions, claims, losses or damages directly caused by such failure to exercise the said skill and care, and shall not include liability for any actions, claims, losses or damages arising out of circumstances incidental or indirectly consequential to such failure.

(c) In addition to any liability Associated Consultant may have under Section 5.07 (b), Associated Consultant shall, at its own cost and expense, upon request of the Consultant, re-perform the Services in the event of Associated Consultant's failure to exercise the skill and care required under Section 5.01(a).

(d) Anything in Sections 5.07 (b) or 5.07 (c) to the contrary notwithstanding, the

Associated Consultant shall have no liability whatsoever for actions, claims, losses or damages occasioned by (i) Consultant's overriding a decision or recommendation of the Associated Consultant or requiring the Associated Consultant to implement a decision or recommendation with which the Associated Consultant does not agree, or (ii) the improper execution of the Associated Consultant's instructions by agents, employees or independent contractors of the Consultant.

5.08 ***Laws and Regulations of the Islamic Republic of Uzbekistan***

The Associated Consultant shall respect and abide by all applicable laws and regulations in Republic of Uzbekistan.

5.09 ***Proprietary Right of the Client in Equipment***

Equipment supplied by the Consultant or the Client for the Project shall remain at all times the property of the Client and shall be returned by the Associated Consultant in accordance with procedures to be determined by the Consultant.

5.10 ***Proprietary Rights of the Consultant in Reports and Records***

All reports and relevant data such as maps, diagrams, plans, statistics and supporting records or materials complied or prepared in the course of the Services shall be confidential and shall be the absolute property of the Consultant.

The Associated Consultant agrees to deliver all these materials to the Consultant upon completion of this Contract. The Associated Consultant may retain a copy of such data but shall not use the same for purposes unrelated to this Contract without prior written approval of the Consultant or the Client.

5.11 ***Insurance***

(a) The Associated Consultant shall take out and maintain adequate professional liability insurance as well as adequate insurance against third party liability.

(b) The Consultant undertakes no responsibility in respect of any life, health, accident, travel and other insurance which may be necessary or desirable for the Personnel of or sub-contractors and specialists associated with the Associated Consultant for the purposes of the Services, nor for any members of any family of any such person.

5.12 ***Reports***

All reports and recommendations and general correspondence from the Associated Consultant to the Consultant under this Agreement shall be in the English language and/or the language requested by the Client.

ARTICLE VI General Provisions

6.01 ***Suspension***

If any of the following events shall have happened and be continuing, the Consultant may by fourteen (14) days prior written notice to the Associated Consultant suspend in whole or in part payments due thereafter to the Associated Consultant under the Agreement:

(a) The Client or the ADB has suspended payments for the Project;

(b) A critical default has occurred on the part of the Associated Consultant in the execution of the Agreement;

(c) Any other condition has arisen which, in the reasonable opinion of the Consultant, interferes, or threatens to interfere, with the successful carrying out of the Project or the accomplishment of the purposes of the Agreement.

6.02 ***Termination of the Agreement by the Consultant***

(a) If any of the following events shall have happened and be continuing, the Consultant may by written notice to the Associated Consultant terminate the Agreement:

(1) Any of the conditions referred to in Article 6.01 continue for a period of fourteen (14) days after the Consultant has written notice to the Associated Consultant of suspension of payments to the Associated Consultant under the Agreement.

(2) The Loan Agreement has terminated in accordance with its terms.

(b) In the event of the Clause 6.02 (a), the Consultant may terminate the Agreement at any time by giving not less than thirty (30) days prior written notice to the Associated Consultant."

6.03 ***Termination of the Agreement by the Associated Consultant***

The Associated Consultant shall promptly notify the Consultant in writing of any situation or of the occurrence of any event beyond the reasonable control of the Associated Consultant which makes it impossible for the Associated Consultant to carry out its obligations hereunder. Upon confirmation in writing by the Consultant of the existence of any such situation or event, or upon failure of the Consultant to respond to such notice within fifteen (15) days of receipt thereof, the Associated Consultant shall be relieved from all liability from the date of such receipt or failure to carry out such obligations, and the Associated Consultant may thereupon terminate the Agreement by giving not less than thirty (30) days prior written notice thereof.

6.04 ***Termination Procedure***

(a) Upon termination of the Agreement under Section 6.02 (a), receipt of notice of termination under Section 6.02 (b) or the giving of notice of termination under Section 6.03, the Associated Consultant shall take immediate steps to terminate the Services in a prompt and orderly manner and reduce losses and to keep further expenditures to a minimum.

(b) Upon termination of the Agreement (unless such termination shall have been occasioned by the default of the Associated Consultant), the Associated Consultant shall be entitled to be reimbursed in full for such costs as shall have been duly incurred prior to the date of such termination and for reasonable costs incident to the orderly termination of the services, and the return travel of the personnel of the Associated Consultant, but shall be entitled to receive no other or further payment.

6.05 ***Settlement of Disputes***

Any dispute or difference arising out of this Agreement or in connection therewith which cannot be amicably settled between the parties shall be finally settled under the Rules of Conciliation and Arbitration of the International Chamber of Commerce by one or more arbitrators appointed in accordance with the said Rules. The arbitration shall take place in the Republic of Korea. The resulting award shall be final and binding on the parties and shall be in lieu of any other remedy.

6.06 ***Force Majeure***

(a) If either Party is temporarily unable by reason of force majeure or the laws or regulations of Uzbekistan to meet any of its obligations under the Agreement, and if such Party gives to the other Party written notice of the event within fourteen (14) days after its occurrence, such obligations of the Party as it is unable to perform by reason of the event shall be suspended for as long as the inability continues.

(b) Neither Party shall be liable to the other Party for loss or damage sustained by such other Party arising from any event referred to in Section 6.06 (a) or delays arising from such event.

(c) The term "force majeure", as employed herein shall mean acts of God, strikes, lock-outs, or other industrial disturbances, acts of the public enemy, wards, blockades, insurrection, riots, epidemics, landslides, earthquakes, storms, lightning, floods, washouts, civil disturbances, explosions, and any other similar events, not within the control of either party and which by the exercise of due diligence neither party is able to overcome.

(d) Any "force majeure" ruling applied under the Client/Consultant Agreement will automatically be applicable to this Agreement.

6.07 ***Variation of Agreement***

The Agreement may be varied by agreement between the Parties. All such variations, including variations in the cost estimates and in the amount specified in Section 3.01 (b), shall be in writing signed by the duly authorized representatives of the Parties.

ARTICLE VII Effective Date, Miscellaneous

7.01 ***Effectiveness***

The Agreement shall become effective upon the date notice is given to proceed with the Services under Section 1.02 and shall be in full force until the Services and all payments therefore have been completed, and at such time the Parties hereto shall be mutually released from all obligations hereunder.

7.02 ***Authorized Representative***

Any action required or permitted to be taken, and any document required or permitted to be executed, under this Agreement may be taken or executed on behalf of the Associated Consultant by Lee Chang Beum, Director General or a designated representative and on behalf of the Consultant by Lee, Sang Ik, Vice President/Division Head of Overseas Projects Division, and/or his designated representative, Cheong Byeong-Ho, the Project Team Leader.

7.03 ***Notices or Requests***

Any notice or request required or permitted to be given or made under this Contract shall be in writing in the English language. Such notice or request shall be deemed to be duly given or made when it shall have been delivered by hand, mail or cable to the party to which it is required to be given or made at such Party's address specified below, or at such other address as either Party may specify in writing.

For the Consultant:

Name: Lee, Snag Ik
Vice President/Division Head of Overseas Projects Division

Address: Dongshin Engineering & Consultants Co., Ltd.
181-1, Jamsil-dong, Songpa-gu, Seoul, Korea

Telephone: +82 2 424 0560
Facsimile: +82 2 424 0002
Email: bigman03@naver.com

For the Associated Consultant:

Name: Lee Chang Beum
Director General, Overseas Project Office
Address: Korea Rural Community Corporation
30 Anyang Pangyo-ro(487 Poil-Dong), Uiwang-si, Gyeonggi-Do, Republic of Korea

Telephone: +82 31 420 3060
Facsimile: +82 31 423 8953
Email: cblee8303@hanmaill.net

IN WITNESS WHEREOF, the Parties hereof have caused the Contract to be signed in their respective names of the day and year first above written.

FOR AND ON BEHALF OF The Consultant	FOR AND ON BEHALF OF The Associated Consultant
Lee Sang Ik Vice President Dongshin Engineering & Consultants Co., Ltd.	Lee Chang Beum Director General Korea Rural Community Corporation.

<u>List of Appendices</u>

- ***A. Background Information and Terms of Reference***
- ***B. Personnel***
- ***C. Estimate of Fee Unit Costs and Out of Pocket Expenses***
- ***D. Staffing Schedule***

APPENDIX A

BACKGROUND INFORMATION AND TERMS OF REFERENCE

DESCRIPTION OF SERVICES
Terms of Reference
Water Resources Management Sector Project

1. BACKGROUND

GENERAL

The Republic of Uzbekistan has applied for a loan from the Asian Development Bank (ADB) in the amount of US$100 million towards the cost of the Water Resources Management Sector Project, and it intends to apply a portion of the proceeds of this loan to eligible payments under contract for engagement of consultants to assist the Ministry of Agriculture and Water Resources of the Republic of Uzbekistan in the implementation of the Project.

PROJECT AREA

The Water Resources Management Sector Project will provide funding for the rehabilitation of irrigation and drainage schemes in the Fergana Valley and the Zarafshan River Basin. Six to eight schemes will be rehabilitated under the Project. Two subprojects, Narpay in the Zarafshan River Basin and Besharyk in the Fergana Valley, have been identified and full feasibility studies completed under the project preparatory technical assistance (PPTA). The remaining subprojects are to be identified and analyzed during project implementation.

The total area to be rehabilitated by the Project will be about 60,000 ha. Subprojects financed subsequent to the Narpay and Besharyk core subprojects will be screened by the PMO and selected based on agreed technical, economic, social, and environmental criteria. Subproject screening will (i) ensure compliance with safeguards, and (ii) provide necessary reconnaissance level studies to confirm remaining criteria.

IMPACT AND OUTCOME

Project benefits include (i) reduced production losses, (ii) increased energy efficiency, (iii) improved water management, (iv) decreased land degradation caused by soil salinity and waterlogging, (v) increased irrigation efficiency, (vi) enhanced farm productivity through increased yields and cropping intensity, and (vii) increased rural livelihoods.

OUTPUTS AND COMPONENTS

To ensure such an outcome, the Project and the accompanying TA are designed to deliver outputs, covering policy, technical and institutional changes. These outputs are: (i) Rehabilitation and upgrading of irrigation and drainage infrastructure, (ii) Improved water resources management, (iii) operational and effective project management and monitoring systems.

The Project comprises the following components:

(i) **Rehabilitation and Upgrading of Irrigation and Drainage Infrastructure**. Project investments will focus on pumping facilities; gravity head-works; inter-farm distribution canals and structures; and inter-farm and on-farm drainage to increase the reliability of irrigation supplies, improve water distribution, and reduce soil degradation. The R&U of pumping facilities and replacement of pumps with more efficient equipment will result in energy savings. Indicative types of R&U include (i)

pumping stations comprising replacement and/or repair of intake works, including bulkhead gates and trash racks; replacement of pumps and motors; upgrading electrical control works; replacement of problematic and/or unreliable reaches of the rising mains; replacement of control valves; provision of appropriate surge protection systems; and repair or replacement of auxiliary works such as floor drains, overhead cranes, and ventilation systems; (ii) improving either major or minor surface water intakes for both pump and gravity systems to provide greater capture efficiencies and improved control; (iii) improving control structures and gauging stations on inter-farm canals consistent with enhanced water management targets; (iv) upgrading and/or replacing gated outlets to on-farm canal outlets from the inter-farm canal and providing a means for measuring the discharge at each outlet point; (v) providing necessary rationalization of both alignment and grade of the inter-farm canals; (vi) providing adequate cross drainage treatment for the inter-farm canals; (vii) providing auxiliary facilities associated with the inter-farm canals such as vehicle and pedestrian bridges and canal access for non-irrigation purposes; and (viii) agricultural drainage facilities, especially in subproject areas affected by high water table and/or levels of soil salinity.

(ii) **Improved Water Resources Management**. The Project will support improved irrigation, drainage, and water resources management from the basin to the on-farm level. The Project will also support a basin and irrigation and drainage management institutional optimization and coordination study to prepare a plan and capacity development program to improve water use efficiency and productivity for inter-farm irrigation and drainage in the Fergana & Zarafshan Vallies. The study will (i) analyze existing institutional and organization structures of BISAs, ISA, MCA, Department of Pump Station Operations (DPSO), WUAs, local government, and other relevant stakeholders; (ii) analyze their existing management rules, regulations, procedures, and systems to identify bottlenecks and inefficiencies for irrigation, drainage, and water management; (iii) analyze coordination between water management stakeholders; (iv) develop options and a plan to improve (a) the institutional and organization structures, (b) management and operational procedures and systems, and (c) coordination between stakeholders; and (v) develop and implement a capacity development program to support the recommendations.

(iii) **Project Management**. The Project will provide implementation support to the project management office (PMO), which established in Tashkent. This will be supported by two project implementation units (PIUs)–one in Fergana city and one in Samarkand city. The PMO will deal with technical issues, procurement, accounting, M&E, water resources management, and safeguards. The PMO and PIUs will have national staff supported by national and international specialists. Comprehensive results-based monitoring will be a critical function of the PMO.

THE SWISS AGENCY FOR DEVELOPMENT AND COOPERATION CO-FINANCING

The joint venture of IWMI and ICWC SIC will be recruited for WUA training and support services. The joint venture has been involved in providing such services to WUAs for the past 5 years under programs financed by SDC. Based on this continuing experience, IWMI and ICWC SIC have developed an in-depth understanding of Uzbekistan's WUAs and the

challenges they face that uniquely qualifies IWMI and ICWC SIC to provide training tailored to their specific requirements. IWMI and SIC are currently carrying out WUA training and other water management activities in the Fergana Valley for SDC, and extending these to the subproject areas to be rehabilitated under the loan is a natural extension of their ongoing activities. Using IWMI and ICWC SIC for WUA training will help to make uniform and promote stronger institutionalization of these efforts throughout the country rather than having a piecemeal approach with different contractors. SDC initially contracted IWMI and ICWC SIC through single source selection because of concern over the lack of capacity in the country to perform these services. Since then, SDC has evaluated and found their performance satisfactory and engaged them for further services.

COST ESTIMATES

The investment cost of the Project is estimated at $144.0 million, including taxes and duties, physical and price contingencies, interest charges during implementation, and commitment charges. ADB will provide a blend of two loans totaling $100 million: (i) $85 million from ADB's ordinary capital resources (OCR), and (ii) an Asian Development Fund (ADF) loan in various currencies equivalent to Special Drawing Rights 10.042 million (estimated at $15 million).

IMPLEMENTATION ARRANGEMENTS

(i) Project Management

The Project will be implemented over 6 years starting in 2009. A project steering committee will guide policy and strategic direction. A project director is appointed by MAWR. A project management office established within MAWR in Tashkent to manage day-to-day implementation activities. Project implementation units established in Samarkand city in the Zarafshan River Basin and Fergana city in the Fergana Valley. A full-time project manager, who will be acceptable to ADB, will manage the project management office, while deputy managers will oversee each project implementation units.

(ii) Implementation Period

The Project will be implemented over 6 years, starting in 2009. The estimated project completion date is 31 December 2014 and the loan closing date is 30 June 2015.

(iii) Procurement

All ADB-financed procurement of goods, works, and consulting services will follow ADB's Procurement Guidelines (2007, as amended from time to time). Major contracts will include (i) supply and installation of pump and motors, and rehabilitation of the pumping stations and associated infrastructure; (ii) civil works for rehabilitation of the inter-farm canals, agricultural drains, and associated infrastructure; and (iii) project management and training support. Civil works contracts costing more than $1 million and contracts for supply (and installation) of equipment and materials valued at more than $1 million, will be awarded under international competitive bidding while those valued at $1 million or less will be awarded following national competitive bidding procedures acceptable to ADB. Equipment and materials packages valued at less than $100,000 will be procured through the shopping mode. The procurement plan will be updated annually to cover the next 18 months. The relevant sections of ADB's Anticorruption Policy (1998, as amended to date) will be included in all bid documents issued during implementation of the Project.

(iv) Consulting Services

The Project will provide 146 person-months of international consultants and 440 person-months of national consultants. Consulting services will include project implementation for components 1 and 2 and support to the PMO and PIUs, including planning, design, and construction supervision of rehabilitation and upgrading works as well as project management, procurement, financial management, institutional analysis, and M&E–most of which will be provided under a single contract package.23 The international and national consultants will be recruited in accordance with ADB's Guidelines on the Use of Consultants (2007, as amended from time to time) using consulting firms, full technical proposals, and quality- and cost-based selection with 80:20 weighting.

(v) Disbursement Arrangements

The proceeds of the ADB loan will be disbursed according to ADB's Loan Disbursement Handbook (2007, as amended from time to time). For consulting services and equipment following international competitive bidding procedures, loan funds will be disbursed through direct payment. For civil works following national competitive bidding procedures and small expenditures related to the PMO, reimbursement and imprest fund procedures will be applied. After loan effectiveness, MAWR will open two imprest accounts, one for each loan at a commercial bank in Uzbekistan acceptable to ADB. These imprest accounts will be established and managed according to ADB's Loan Disbursement Handbook. Total outstanding advances are not to exceed the estimated share of eligible project expenditures to be financed through the imprest account for the next 6 months or 10% of the loan amount, whichever is lower. ADB's statement of expenditures procedure will be applied when reimbursing eligible project outlays and liquidating and replenishing the imprest accounts for individual payments not exceeding $100,000.

(vi) Accounting, Auditing, and Reporting

MAWR, through the PMO, will establish financial management and accounting systems for the Project in accordance with the Guidelines for the Financial Governance and Management of Investment Projects Financed by the ADB (2002). The PMO will also maintain separate records and accounts for goods, works, and services to be financed out of the loan proceeds following sound accounting principles. Independent auditors acceptable to ADB will audit these accounts and records yearly. Within 6 months of the close of the financial year, certified copies of the audit report, together with the auditor's opinion, will be submitted to the Government and ADB in English. The audit reports will include a management letter and a separate opinion on the use of the imprest accounts and statement of expenditures procedure. For the external audit of accounts, the loan proceeds may finance the services of private auditors and the translation of their reports into English. The auditors are to be recruited in a manner acceptable to ADB.

MAWR, through the PMO, will submit quarterly and annual reports to ADB. The reports will indicate progress made, problems encountered, steps taken to remedy the problems, and a program of activities–along with expected progress during the remainder of the implementation period. The reports will incorporate project performance monitoring data and all relevant financial data. The reporting system will focus on outcomes, efficiency, and quality; and will be consistent with international and local reporting standards, as well as with the design and monitoring framework (Appendix 2). MAWR will also provide other reports and information relating to the Project as ADB may reasonably request. Within 6 months of project completion, MAWR will submit a project completion report to ADB that includes information on project implementation, use of both the ADF and OCR loan

proceeds, and accomplishments in relation to the Project's outcome and impact.

(vii) Project Performance Monitoring and Evaluation

The PMO will establish a project performance and monitoring system to monitor the Project's progress in achieving the planned outputs, outcome, and impact within 6 months of loan effectiveness. An M&E unit will be established in the PMO to measure results and report findings of the project progress to the Government and ADB. A matrix of indicators will be established and baseline data for the agreed indicators will be gathered within 6 months of loan effectiveness. The indicator data will be updated every 6 months and reported to stakeholders and the public. During loan review missions, the indicators will be validated and may be modified to reflect project performance and results more accurately.

(viii) Project Review

MAWR and ADB will jointly conduct periodic reviews to assess (i) physical, institutional, and capacity development progress; (ii) financial progress; (iii) implementation arrangements; and (iv) identify and resolve problems. The periodic reviews will initially be conducted on a semiannual basis. Starting from the second year of the Project's implementation and if deemed appropriate by ADB, the periodic review may be conducted on an annual basis. A comprehensive midterm review of the Project will be carried out immediately following physical completion of the R&U of the Narpay and Besharyk core subprojects and completion of the final design for the batch 3 subprojects (expected to be toward the end of year 3 according to the implementation schedule). The midterm review will evaluate (i) project scope, design, and implementation arrangements; (ii) progress against agreed indicators, including the design and monitoring framework; (iii) procurement performance; (iv) performance of the consultants; and (v) any other parameter agreed by MAWR and ADB. At the conclusion of the midterm review, ADB and MAWR may agree on changes in both project scope and implementation arrangements, as deemed necessary.

(ix) Anticorruption Measures

ADB's Anticorruption Policy (1998, as amended to date) was explained to and discussed with the Government and MAWR. Consistent with its commitment to good governance, accountability, and transparency, ADB reserves the right to investigate, directly or through its agents, any alleged corrupt, fraudulent, collusive, or coercive practices relating to the Project. To support these efforts, relevant provisions of ADB's Anticorruption Policy are included in the loan regulations and the bidding documents for the Project. In particular, all contracts financed by ADB in connection with the Project shall include provisions specifying the right of ADB to audit and examine the records and accounts of the Executing Agency and all contractors, suppliers, consultants, and other service providers as they relate to the Project.

In relation to the Project, the Government will ensure the disclosure of all project-related procurement actions, including consulting services contracts and the procurement of major equipment and civil works, in local newspapers and on the internet. Such disclosed information will include, among other matters, the list of participating bidders; the winning bidder; a summary of the tender procedures; the value of the contract awarded; and the general terms and specifications of the goods, works, or services procured. The financial management capacity of the Executing Agency has been assessed as satisfactory. The project civil works and equipment procurement have been packaged into a large contract to attract reputable contractors and will require ADB's approval before being awarded.

2. THE OBJECTIVE

The Consultants will assist PMO under Ministry of Agriculture and Water Resources in implementation of the following major project components:

(i) Rehabilitation and upgrading of irrigation and drainage infrastructure;
(ii) Improved water resources management;

Project Management.

3. THE SCOPE OF THE CONSULTING SERVICES

The services will be provided by an international consulting firm in association with domestic consultants and/or domestic consulting firm(s). The consultants will work closely with the Project Management Office (PMO) based in Tashkent and with the Project Implementation Units (PIUs) in Fergana and Samarkand. The project will require 146 person-months of international consulting services and 440 person-months of national consulting services to assist the PMO under Ministry of Agriculture and Water Resources (MAWR) in managing and implementing the Project. The project will be implemented by 31 December 2014.

The Consultant team as a whole shall be jointly liable and responsible for the 100% execution and the completion of the following described tasks, whether such tasks are described under different components/groups, and/or particular positions.

The general scope of the services required to be provided under the Project will include, but not be limited to:

(i). assist the PMO in reconnaissance review of candidate projects to ensure that they are prima facie, in conformance with the agreed selection criteria;
(ii). undertake necessary feasibility level studies, institutional analysis, environmental assessment, resettlement considerations and preliminary design necessary to establish the economic viability of investment in integrated irrigation infrastructure;
(iii). prepare feasibility level outputs documenting the viability of the project works while adequately addressing both environmental and resettlement issues and outlining in detail the implementation arrangements and monitoring and evaluation (M&E) procedures;
(iv). parallel to the preparation of the proposed Subproject's feasibility study and for the core Subprojects, carry out required additional surveys, inculding geological, hydrological and ultra-sound examination of the rising mains;
(v). agree the prepared feasibility studies with all the ministries and agencies, and submit them to the Government of the Republic of Uzbekistan for approval;
(vi). conduct a detailed hydraulic analysis of the pumping system including the rising mains and prepare specifications and bid documents for the pumps and motors;
(vii). prepare detailed designs for all features associated with the pumping plant including, but not limited to pump suction and rising mains including common discharge header, surge protection, drainage and suction pumps and electrical equipment, as well as control and protection systems;
(viii). analyze design options for all aspects of the head-works (if applicable), the pumping plant and inter-farm canal rehabilitation and upgrading including cross and head regulators, on-farm canal outlets and associated flow measurement provisions, bifurcation structures, measurement structures, lining repair and replacement including all necessary earthworks, escapes, bridges as

well as evaluating potential construction modalities to be considered with a view to timely and cost effective rehabilitation;

(ix). evaluate all design options for the main and branch inter-farm canals including associated measurement and control structures as well as on-farm canal outlets in conjunction with the concerned water users' association (WUA), reaching agreement on the package of improvements for the relevant works;

(x). finalize the detailed design, construction drawings, bills of quantities, technical specifications and tender documentation for all aspects of each bid package and agree them with all relevant local administrations and authorities;

(xi). provide construction supervision and necessary quality control testing for all installation and construction work[1], keep daily records of completed works and make notes on works completed by the contractors to the appropriate record books;

(xii). assist PMO in overseeing the work of the service contractors providing support to WUAs in the Subproject areas and in carrying out an irrigation and drainage institutional optimization study for the Fergana & Zarafshan valleys formulating recommendations for required strengthening and providing necessary institutional based training to Basin Irrigation System Administration (BISA), Irrigation System Administration (ISA) and Main Canal Administration (MCA) staff in the Valley.

(xiii). support PMO and the two PIUs in M&E of all aspects of project implementation and initial operations.

The Consultant shall carry out such other activities, which may not be mentioned above, required for the due completion of his assignment, and prepare on behalf of the working committee the acts of completion of each work cycle, including the acts of the state committee, and participates in handing-over of the completed works to the State committee.

The scope of the consultant assignment is divided as follows:

A. Rehabilitation and upgrading of irrigation and drainage infrastructure

The consultants performing the services under this package will documents, and will supervise reconstruction and rehabilitation of the Narpay Machine Canal and Besharyk Lift scheme; will prepare feasibility study and detailed designs, tender documents for Batch 2 under the proposed sector loan. Project studies for six additional Subprojects will initially be at full feasibility level and upon establishment of economic viability; these Subprojects as well as Narpay and Besharyk will proceed to detailed design and tender document preparation. The design activities, preparation of the work plan and operations schedule for rehabilitation and upgrading of the Narpay Machine Canal and the Besharyk Lift scheme will be carried out in parallel with the preparation of the documents for the subprojects in Batch 2. Contract award and the initial phases of construction supervision related to the works for Narpay and Besharyk will parallel detailed design of Batch 2. This approach will ensure timely utilization of the proceeds of sector loan and minimize the commitment charges incurred by the Government of Uzbekistan while at the same time ensuring improved agricultural production from the command areas of the selected Subprojects at the earliest possible date.

[1] With respect to construction supervision, the consultants will act as "engineer" under the terms of the Federation International des Ingenieurs-Conseils (FIDIC) conditions of contract.

B. Project Management

The consultants shall undertake the following tasks under this component:

(i) *Project Management*

(a) Assist PMO in coordinating all matters related to project implementation with relevant ministries of the Government of Uzbekistan (MAWR, Ministry of Finance, State Committee on Nature Protection and others), PIU, design institutes, the consultants hired under the technical assistance associated with the Project, the local administration, ADB, and other organizations related to project implementation;

(b) Provide EA staff with on-the-job training in implementing the ADB guidelines and procedures, efficient project management and scheduling techniques;

(c) Assist PMO in programming project activities, estimating the financial requirements for these activities and the release of funds on time;

(d) Assist the PMO in coordination with PIU and other agencies to ensure that the tendering procedures for procurement of works, services, equipment, and materials are in line with ADB procedures, and that all steps are taken expeditiously and in a transparent manner;

(e) Assist the PMO to ensure that the accounting standards for the Project meet ADB requirements and that the withdrawal applications sent to ADB are complete and are sent on a timely basis;

(f) Assist the PMO to ensure that all periodic reports are prepared systematically, submitted on time, reflect the real picture of project implementation; that major issues relating to project implementation are brought to the attention of the concerned parties; and that necessary remedial measures are implemented;

(g) Provide co-ordination the activity of all consultants and sub-consultants within this ToR and Project as well as timely and efficiently execution by them their personal tasks;

(h) Familiarize the Project staff with procurement requirements of the Project, including specific ADB guidelines and procedures on procurement and use of consulting services. Develop contracting capacity at the EA, paying particular attention to on-the-job transfer of knowledge, as well as organizational and management considerations;

(i) Based on the procurement requirements of the Project components, prepare detailed procurement plans and packages, and determine realistic time-bound schedules for procurement, including parallel and sequential steps for completing procurement activities from initial planning to delivery of goods and services;

(j) Provide advice and assistance in organizing and managing procurement activities, and review the appropriateness of various procurement packages with respect to their methods of procurement, to ensure that procurement can be undertaken in the most efficient and cost-effective manner in accordance with ADB guidelines applicable to the Project;

(k) Provide advice and guidance in connection with the application of procedures and requirements related to prequalification, bid opening, evaluation and award of contracts, contract registration in authorized agencies, custom clearance of the goods and they delivery to the final recipients. Particular attention should be paid to time allowance for preparation of bids, bid opening procedures, circumstances related to clarifications of bids, confidentiality, examination of bids, evaluation and comparison of bids, and applicable domestic preferences;

(l) Interpretation of various contract provisions and settlement of disputes with the contractors as necessary;

(m) Establish a procurement monitoring system, to allow collection and recording of procurement data for: (a) timely information flow, submission and approval of terms of reference, shortlists, and other requirements; (b) tracking all necessary and critical procurement actions and activities; (c) prompt reporting of contract award information by the EA to ADB; (d) preparation of quarterly reports to ADB; and (e) revision of timing of procurement actions including advertising, bidding, contract award, and completion time for individual contracts. This procurement monitoring system will be maintained and kept up-to-date by the PMO. The PMO Staff will be trained by the Consultant to use this system;

(n) In relation to contracts, develop administrative and technical procedures for review of contract performance as an integral part of the approval process for justifying contract payments; and

(o) Develop a contract management system to ensure that records and data are stored systematically, and cross-referenced with the financial accounts of the Project. The system must allow safe keeping of procurement documentation for easy retrieval and referencing, with adequate paper trails in accordance with generally accepted standards. The contract management system must also be designed to support loan disbursement, reporting, and accounting requirements of the Project. Familiarize the PMO staff with the documentation requirements of loan disbursement applications, as well as the necessary records that must be kept by the project to support these loan disbursement applications.

(ii) *Project Implementation Monitoring & Evaluation*

a. Develop a project monitoring and evaluation (M&E) system, including (i) long-term environmental and social impact monitoring programs; and (ii) quantifiable indicators to monitor and measure the performance, level of maintenance and efficiency of the rehabilitated irrigation and drainage system;

b. Ensure that such Evaluation system will be compatible or improve the existing Government's system to monitor development progress in the agricultural sector so that the project impact monitoring can be performed effectively by the Government after Project completion;

c. Develop a standard format for reports that will clear reflect project progress in the context of each its components;

d. Ensure the focus of impact evaluation of the Project on the appropriate design of baseline, midterm, and project completion surveys, including the impacts of agricultural policy reforms on the household standard of living as well as Government budgetary resources;

e. Prepare manuals and conduct training for PMO and PIU staff (monitoring and evaluation unit/MEU) to conduct regular monitoring of the Project activities and prepare project progress reports.

TECHNICAL STUDIES AND SURVEYS

Technical studies and surveys, engineering and geotechnical surveys, including ultra-sound examination of rising mains, drawings, social surveys, material testing, quality control testing, and land acquisition and resettlement plans preparation has been included in the project to support the consultants with their contractual work. The above services will be executed by subcontractors hired by the PMO in close consultation with the Consultant. The Consultant will develop the TORs, technical specification and provide implementation supervision of the said services.

INSTITUTIONAL ARRANGEMENTS

Organizational Structure

MAWR is the Executing Agency and have overall responsibility for the Project. A MAWR staff member is appointed Project director to oversee project implementation. MAWR had established, prior to disbursement of loan proceeds, the PMO in Tashkent to manage project implementation. To support PMO activities in the field and coordinate local activities for implementation of subprojects, prior to disbursement of loan proceeds, PIUs have been established in the Syrdarya-Sokh BISA in Fergana city and in the Zarafshan BISA in Samarkand city.

The Consultants will report to the Project Manager of PMO under MAWR.

To implement the tasks assigned under the terms of reference, the consultants will work with Project Manager as well as with experts and specialists of PMO and PIUs.

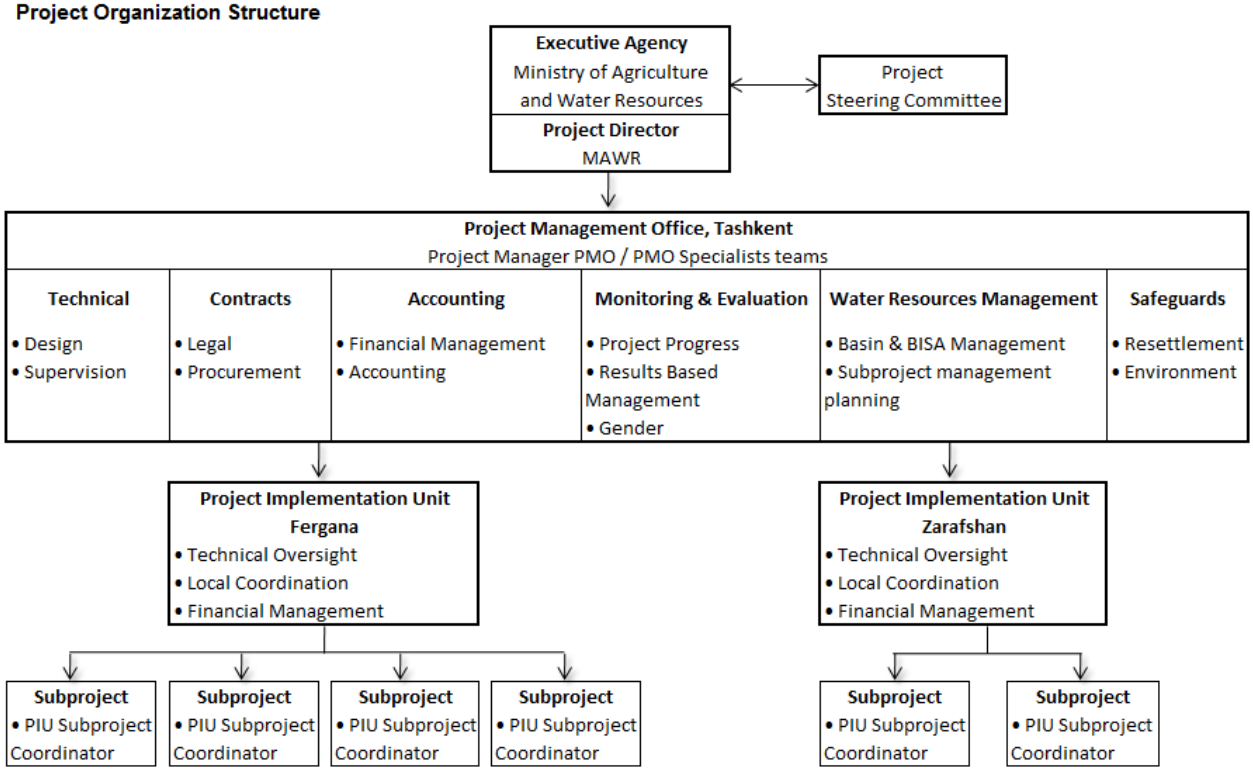
Annex 1

CONSULTING SERVICES
Preparation of Subsequent Sub-Projects, Detailed Design and Construction Supervision of All Sub-Projects and Operational Support for PMO

	Person Month Input per Year					
Specialist	Year 1	Year 2	Year 3	Year 4	Year 5	**Total**
Water Resources Engineer/Team Leader	6.0	11.0	11.0	8.0	8.0	44.0
Pumping Plant Specialist	6.0	8.0	5.0	4.0	2.0	25.0
Electrical Engineer	2.0	2.0	2.0			6.0
Irrigation/Drainage Engineer	6.0	8.0	4.0	4.0	2.0	24.0
Procurement Specialist	3.0	3.0	3.0			9.0
Project Economist	3.0	3.0			3.0	9.0
Monitoring and Evaluation Specialist	2.0	1.0	1.0			4.0
Environmental Specialist	2.0	2.0	1.0			5.0
Water Management Specialist	3.0	8.0	7.0		2.0	20.0
Total International Input (p/m)						**146.0**
Deputy Team Leader (Zarafshan)	7.0	11.0	11.0	7.0		36.0
Deputy Team Leader (Fergana)	7.0	11.0	11.0	10.0	11.0	50.0
Pumping Plant Specialist	7.0	11.0	11.0	11.0	11.0	51.0
Electro/Mechanical Engineer	6.0	8.0	4.0	4.0	4.0	26.0
Irrigation/Drainage Planner	3.0	11.0	3.0			17.0
Agronomist	3.0	9.0				12.0
Resettlement Specialist	3.0	11.0	3.0			17.0
Monitoring and Evaluation Specialist	6.0	11.0	11.0	11.0	11.0	50.0
Gender Specialist	3.0	10.0				13.0
Environmentalist	3.0	9.0	5.0			17.0
Irrigation/Drainage/Design Engineer	3.0	11.0	11.0	3.0		28.0
Procurement Specialist	3.0	11.0	11.0			25.0
Construction Engineers (2)		8.0	22.0	22.0	22.0	74.0
Water Management Specialist	3.0	6.0	6.0	6.0	3.0	24.0
Total National Input (p/m)						**440.0**

Annex 2

Project Organization Structure



BISA = basin irrigation system administration, MAWR = Ministry of Agriculture and Water Resources, PIU = project implementation unit, PMO = project management office, WUA = water users' association

Annex 3. Terms of Reference for Key Individual Specialists[2]

Water Resources Engineer/Team Leader – International
Duration: 44 months
Qualifications: The Team Leader will have at least a Master's degree in civil/irrigation engineering and a minimum of twenty years relevant experience in the planning and design and construction of lift irrigation schemes and irrigation infrastructure. At least five years of that experience shall have been as team leader and at least five years will have been on projects in South or Central Asia.
TOR: (i) Provide overall direction to all specialists making up the consulting team and appropriately group individuals into work units responsible for a particular feasibility study, detailed design undertaking and/or construction supervision activities; (ii) Manage relationships with PMO, MAWR, relevant BISAs, the Uzbekistan Government and ADB as well as with other stakeholders including farmers and WUAs; (iii) Ensure timely delivery and quality control of all required outputs including feasibility studies for rehabilitation and upgrading of the remaining sub-projects, detailed design and contract documents for all sub-projects and as-built drawings of completed works (including Narpay Machine Canal and the Besh Arik Sub-projects); (iv) Prepare detailed, time bound work plans for the preparation, design and tendering of all civil works contract packages envisaged for the works[3] assigning various team members to each key task; (v) Provide technical support and guidance in all aspects of pumping plant, inter-farm canal and drainage system design and tendering; (vi) Coordinate and supervise the preparation of tender documents for rehabilitating and upgrading pumping plants, inter-farm canals and agricultural drainage systems; (vii) In close coordination with PMO, supervise the tendering process and subsequent evaluation of bids and preparation of award recommendations for all contracts for rehabilitation and upgrading; (viii) Develop the TORs, technical specification and provide implementation supervision for technical studies and surveys, including engineering and geotechnical surveys and drawings, social surveys, material testing, quality control testing, as-built surveys, and land acquisition and resettlement plans preparation, and other services that will be procured by the PMO; (ix) Provide coordination and oversight to ensure that monitoring and evaluation, resettlement, environmental and water management support aspects of all sub-projects are adequately addressed at all stages of project implementation; and (x) Monitor the progress of all planning and design work ensuring that deadlines relating to delivery dates are met.

Pumping Plant Specialist – International
Duration: 25 months
Qualifications: The Pumping Plant Specialist shall have a Masters Degree in Mechanical Engineering and at least fifteen years experience in the planning, design and construction of irrigation pumping plants. At least five years of the experience will be in respect to irrigation pumping plants in Central or South Asia.

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[2] Terms of Reference for individual specialists should be read in conjunction with the capsule TOR for the undertaking as a whole. Some specialists will be involved in only the feasibility level studies while others will be involved in detailed design and/or construction supervision.
[3] These include among others, supply and install contracts for rehabilitation and upgrading of the Narpay and Besh Arik pumping plants, civil works contracts for the Narpay and Besh Arik inter-farm canals and civil works contract for drain rehabilitation within the Besh Arik irrigated command.

TOR: (i) During the screening of additional sub-projects, visit the sites of the subprojects for assessment of present condition of pumps, motors and associated mechanical and electrical systems and collect required data with assistance from PMO/PIU and the national consultants; (ii) Carry out preliminary assessment and prepare preliminary designs and cost estimates for the rehabilitation work for the purpose of preliminary economic analysis ensuring that the short listed projects will fully meet the agreed selection criteria; (iii) During the feasibility phase visit the site for detailed assessment of the condition of the existing pumps, motors, piping, valves, overhead crane, rising mains and all other equipment relevant to the pumping stations; (iv) Identify all repairs and replacement works necessary for the rehabilitation of Pumping Stations: (v) Prepare a feasibility level design of for each proposed pumping station based on the calculated peak water requirements for the command area, finalize pump parameters and motor ratings, prepare basic drawings such as system flow diagrams, technical description of the works, a preliminary bill of quantities, cost estimates etc. as required for completion of feasibility reports; (vi) Review the design level field information gathered and verify the pump parameters proposed at the feasibility stage, particularly the pumping head, and carryout a detailed hydraulic analysis of the pumping system including the rising mains; (vii) Prepare specifications, data sheets and tender documents for the pumps and motors in coordination with the Electrical Engineer: (viii); Work with the Procurement Specialist and carry out prequalification process for pump and motor supply and install contract[4] for the purpose of international competitive bidding (ICB) for pump station rehabilitation and upgrading; (ix) Design pump suction and discharge piping and prepare layout for realignment of piping outside the pump house including a common discharge header if appropriate and the rising mains;[5] (x) Carry out preliminary analysis of hydraulic surge effects on the pumping system and design a surge protection system adequate to protect the pumping equipment and frame the specifications for the same; (xi) Design an appropriate pump-house ventilation system and prepare associated equipment specifications, data sheets, drawings etc; (xii) Undertake site visits to firm up the repair/ replacement works to intake works, over head crane, pump house building as envisaged at the time of feasibility and finalize work descriptions, quantities, specifications, drawings etc. as required for inclusion in bid documents; (xiii) Prepare a comprehensive bid document for rehabilitation of the pumping system on a supply and install basis including required drawings, data sheets and specifications; (xiv) Assist PMO in providing clarifications to bidder's queries and in the technical evaluation of bids received; (xv) Prepare a comprehensive operation manual for the pumping system and provide familiarization with operational procedures to all appropriate BISA an ISA staff; and (xvi) Provide supervision during the installation and construction of the works and continuing guidance during commissioning whenever possible.

[4] Present thinking is to award a single supply and install contract for each pump-house under international competitive bidding which will include supply, fabrication and installation of all pumps, motors, piping, valves, electrical system as well as the rising mains and associated civil works related to pump foundations and building refurbishment.

[5] For each pump station going to final design a comprehensive testing program to determine the thickness of the rising main conduits will be carried out during the winter closure to provide an adequate basis for specifying those reaches to be replaced.

Electrical Engineer – International

Duration: 6 months

Qualifications: The Electrical Engineer will have a degree in Electrical Engineering and at least 10 years experience in design of power supply for irrigation pumping plants and all associated control circuitry.

TOR: (i) For those projects under feasibility study visit the sites of the sub project for assessment of the present condition of the electrical system and collect required data with the help of National consultants and carry out a preliminary assessment and prepare feasibility level designs and cost estimates; (ii) During the design stage of the sub-projects, including Narpay and Besh Arik among others, visit the site for detail assessment of the existing condition of the electrical system serving the motors driving the pumps; (iii) undertake a brief study of the existing high voltage power supply system and high voltage sub stations supplying power at 6 Kv to the pumping stations with connected loads expected to range from 8 to 14 MW:[6] (iv) Based on the recommended option from (iii) above, prepare technical specifications and data sheets. for electric motors as a part of bid documents; (v) Design all electrical systems for the pumping stations including cable sizing, cable layout, equipment layout in both the switchgear room and control room, lighting and an effective earthing (grounding) system; (vi) Design the control, protection and automation systems as well as the control desk: (vii) Prepare specifications, data sheets, single line diagrams, control schematics, drawings, bills of quantity, and schedule of rates for all aspects of the electrical systems and cost estimates for the same; (viii) Provide clarifications to bidder's queries and assist PMO in technical evaluation of bids; (ix) Participate in supervision of installation and construction as appropriate; (x) Prepare electrical related aspects of the Operation Manuals for each pumping station; and (xi) support PMO during the commissioning period and during initial operations.

Irrigation and Drainage Engineer – International

Duration: 24 months

Qualifications: The Irrigation and Drainage Engineer will have a Master's degree in civil/irrigation engineering and a minimum of fifteen years professional experience in the planning, design and construction of medium and large-scale irrigation systems. At least five years of that experience will have been related to lift irrigation systems in Central and/or South Asia.

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[6] The study will specifically address the following points: (i) Current practice followed in Uzbekistan regarding grid operation, grid power factor control, and measurement of power (energy) consumption at the pumping stations; (ii) Effects on system stability and the system power factor were squirrel cage induction motors (with various starter types including soft starters) are used to replace the synchronous motors currently in use; (iii) identify additional measures and/or equipment required to maintain system stability and the system power factor were squirrel cage induction motors to be used and cost of such measures / equipment; (iv) A financial/economic study of the two alternatives based on use of synchronous motors and squirrel cage induction motors; and (v) Frame recommendations as to the type of motors to be used as replacements for powering the pumps.

TOR: During Feasibility Studies (i) Review feasibility studies earlier prepared for the Narpay and Besh Arik Sub-projects with a view to adoption of best practice innovations on ensuing projects; (ii) Based on both current and future cropping patterns, and determining the crop water requirements for each crop included and based on current and improved future conveyance efficiencies, determine the irrigation requirements at the pumping plant, inter-farm canal head and key junctures along its course, and at each on-farm canal outlet; (iii) Prepare feasibility level designs for rehabilitating and upgrading the inter - farm canal system including all branches and associated control structures and water measurement facilities as well as other required supporting infrastructure including but not limited to cross drainage, emergency escapes and bridges for the commands of each of the remaining sub-projects; (iv) Adopt and recommend innovations into the canal designs that will improve irrigation conveyance and operational efficiency and delivery flexibility ensuring equitable distribution of surface water throughout the command; (v) Review and recommend any changes in supply arrangements from the pumping plant to the inter-farm canals and from the inter-farm canals to on-farm canals to improve either operational efficiency or equity of distribution: (vi) Ensure that adequate flow measurement sites are provided through out the system to facilitate both system management and monitoring of system performance; (vii) identify all rehabilitation/upgrading required on the agricultural drainage system serving the irrigated command and package and cost the drainage rehabilitation work (viii) Work with the local agriculturist in making adequate provision for the water requirements of higher value crops which may be introduced into the cropping pattern in the future; (ix) Prepare feasibility level designs[7] and cost estimates appropriate to establishing the feasibility of the sub-project; During design: (ix) Assume overall responsibility for management and supervision of the design for all irrigation and drainage rehabilitation and upgrading work and timely consultation on all design considerations with MAWR, PMO and ADB; (x) Organize and supervise all topographic, profile and cross-section surveys required[8] to provide necessary input data for detailed design of the inter-farm canals, agricultural drains and associated works; (xi) Undertake the design of rehabilitation and upgrading or replacement works including but not limited to (a) inter-farm canal re-sectioning, grading, lining repair and replacement and rerouting if required, (b) head regulators, (c) cross regulators and on-farm canal outlet replacement/rehabilitation, (d) upgrading and/or replacement of existing water control and bifurcation structures and provision of additional structures as required, (e) cross drainage structures such as inverted siphons and overshoots, (f) required bridges and culverts, and (g) associated agricultural drainage works; (xii) In the design of all structures and features of a repetitive nature, standard designs, pre-approved by the Team Leader and PMO should be used as appropriate to minimize duplicity of design inputs; (xiii) Coordinate and supervise the preparation of tender documents for the rehabilitation and upgrading of all designated works; and during construction; (xiv) Organize and oversee the supervision of construction work related to

[7] Where possible standard designs for structures of a type that will be repetitively used in the sub-project and subsequent sub-projects should be prepared in adequate detail to allow their repeated use in both subsequent feasibility studies and for final design. Standardization, where appropriate, will save considerable time and possible duplication of effort.

[8] Please refer to the Paragraph C “Technical studies and surveys” of the Section 3 “The scope of the consulting services”

irrigation and drainage rehabilitation; and (xv) ascertain the need for and oversee the preparation of necessary as-built drawings.

Procurement Specialist – International

Duration: 9 months

Qualifications: The Procurement and Contracts Specialist shall have at least a BS degree in Civil or Mechanical Engineering with 15 years experience in the procurement of civil and mechanical works and contract management. At least five years of this experience will have been specifically related to procurement under ADB/World Bank Projects using international competitive bidding procedures.

TOR: (i) Prepare detailed procurement plans and packages and determine realistic time bound schedules for procurement, including parallel and sequential steps for procurement of civil and mechanical works from initial steps to the delivery of the services under the contracts for rehabilitation and upgrading work associated with the Narpay Machine Canal and Besh Arik Lift Sub-projects as well as those for all subsequent sub-projects to be included in Batches 2 and 3; (ii) Develop, in consultation with MAWR, PMO and ADB, the pre-qualification criteria, prepare notices of pre-qualification and pre-qualification documents and conduct the pre-qualification of international contractors in accordance with both MAWR procedures and ADB guidelines for applicable contract packages under each of the sub-projects; (iii) Under the direction of the Team Leader and using input from various specialists on the team prepare the international tender documents for the rehabilitation and upgrading of pumping plants at both Narpay and Besh Arik as well as all similar packages of work for subsequent sub-projects in FIDIC format agreeable to MAWR, PMO and ADB;[9] (iv) Organize and supervise appropriate pre-qualification exercises for local civil works contractors for the various civil works packages including both irrigation and drainage rehabilitation to be tendered under all sub-projects using procedures agreeable to MAWR and in accordance with applicable ADB Guidelines; (v) Under the direction of the Team Leader and using input from various specialists on the team prepare the tender documents for national competitive bidding for the rehabilitation and upgrading of irrigation and drainage works at both Narpay and Besh Arik as well as all similar packages of work for subsequent sub-projects in FIDIC format agreeable to MAWR, PMO and ADB; (vi) Invite the pre-qualified bidders to submit bids and advise the committee established for evaluation regarding bid opening and the technicalities and process of evaluation in accordance with ADB's guidelines and MAWR requirements pertaining thereto; (vii) Advise on preparation of the summary of evaluation, recommendation for award and contract negotiations with successful bidders; and (vii) In collaboration with the team leader and the Project Director familiarize PMO and PIU staff with the procurement procedures to be followed under subsequent contracts, including ADB's *Guidelines for Procurement* and undertake other activities targeted to developing the procurement and contracting capacity of MAWR staff.

Project Economist – International
Duration: 9 months

[9] The documents to be prepared under (iii) above will include (a) invitation to bid, (b) instructions to bidders, (c) form of bid, (d) form of contract, (e) special and general conditions of contract, (f) drawings and specifications, (g) bill of quantities, (h) schedule of completion, and (i) all necessary addenda;

Qualifications: The Project Economist will have a Master's degree in Agricultural or Project Economics and at least fifteen years relevant professional experience in costing and analyzing the economics of major irrigation investment projects. At least five years of that experience will have been related to irrigation investments in Central and/or South Asia.

TOR: (i) Using input from various specialists on the team develop detailed feasibility level cost estimates for each of the ensuing projects using COST TAB (version to be subject to prior agreement with ADB) ensuring that all input and output can be used by ADB during loan administration; (ii) From the detailed cost tables identify all costs directly related to generating "with-project" benefits; (iii) Working with the planning engineers and agriculturist, identify and quantify the estimated benefits resulting from the investments including the benefits associated with reduced risk of pump failure and the associated interruptions in irrigation water supply as well as the increases in agricultural productivity (higher cropping intensities and/or increased yields) within the relevant command areas resulting from the improved, more efficient, dependable and equitable delivery of irrigation water to the on-farm outlets, the associated WUA support component and improved agricultural drainage; (iv) Undertake detailed economic analysis of all ensuing – sub projects in accordance with the *ADB Guidelines for Economic Analysis* ensuring that each meets the viability requirements of both ADB and MAWR and is fully compliant with the agreed selection criteria; (v) Prepare with and with-out project farm budgets for typical farms in the command of each sub-project to assess the financial impact on individual farms; (vi) Work with the agriculturist and social specialists to evaluate the linkages between rural poverty in the project areas and irrigated agriculture assessing the impact of the projects on poverty;[10] and (vii) In year 6 of the Project reassess the economic return from the earlier sub-projects to provide input for both MAWR's and ADB's Project Completion Reports.

Monitoring and Evaluation Specialist – International

Duration: 4 months

Qualifications: The Monitoring and Evaluation Specialist will have a Master's Degree in Economics/Social Science and at least fifteen years experience in developing and implementing monitoring and evaluation strategies and plans. Five years of that experience should be on large-scale irrigation projects in South or Central Asia.

TOR: (i) Assist PMO in the development and establishment of an appropriate monitoring and evaluation strategy and plans for rehabilitation and upgrading of pumping plants, inter-farm canal systems and agricultural drainage for all sub-projects including Narpay Machine Canal and the Besh Arik Lift Irrigation Sub-projects; (ii) The monitoring and evaluation systems referred to in (i) above will include input, progress, output and impact indicators and be computerized for each Sub-project to which they apply; (iii) Plan, design and supervise the conducting of baseline, intermediate and end of Project socio-economic surveys in the Narpay, Besh Arik[11] and other selected sub-projects; (iv) Identify a set of social and economic indicators that will serve as the benchmark for assessing the impact on farm income in both head and tail reaches of the inter-farm canals and in saline and non-saline areas if applicable. The indicators identified should also allow assessment of the direct and indirect social impacts of the

[10] This consideration will be principally toward agricultural labor in the command areas and both short and long term employment opportunities generated by the sub-projects.

[11] Full use will be made of the baseline data generated under the PPTA during the feasibility studies for the Narpay and Besh Arik Sub-projects.

Projects on the different categories of beneficiaries and vulnerable groups residing in or adjacent to the impacted irrigated commands; (v) Develop appropriate analytical methodology for the socio-economic impact assessment; (vi) Conduct in depth workshops for MAWR/PMO/PIU staff and for those sub-contractors who will carry out the surveys in the field, covering the survey design, sampling criteria, questionnaire requirements, interview methodology and survey analysis; and (vii) Supervise the work of those conducting the field surveys and analyzing assessment results.

Environmental Specialist – International
Duration: 5 months
Qualifications: The Environmental Specialist will have a Master's degree in Environmental Science with at least fifteen years experience in conducting environmental screening/assessment of major water resources projects. The Environmentalist will be fully familiar with both the Government of Uzbekistan and ADB's Environmental Guidelines and at least five years of his experience should be related to projects in South and/or Central Asia.
TOR: (i) Assess and compare Uzbekistan's environmental legislation and frameworks with those of ADB and best international practice to identify gaps, differences or conflicts and recommend modifications and/or mechanisms to promote compatibility; (ii) Prepare initial environmental examinations (IEEs) for each of the Sub- projects selected subsequent to Narpay and Besh Arik to comply with ADB's environmental assessment guidelines and where applicable the specific requirements for sector type loans described therein; (iii) In cases where required conduct full environmental impact assessments which are in full compliance with ADB guidelines; (iii) Propose any environmental mitigation required and prepare environmental management and monitoring plans therefore and provide full details regarding the estimated costs of the required mitigation; (iv) Assess the capacity of MAWR/PMO/PIUs for environmental assessment and implementation of environmental management plans, and quantify training requirements for capacity building; (v) Develop training materials for PMO/PIU staff to support environmental protection measures and to monitor and mitigate potential environmental impacts; (vi) Assess the environmental capacity of the Uzbekistan government agencies involved in vetting and approving the environmental assessments and develop environmental assessment and review procedures and prepare detailed arrangements, agreeable to all parties, which may be used throughout the sector program to facilitate review of documents related to the various sub-projects; (vii) Work with relevant team members in an effort to provide particular emphasis with regard to environmental management of irrigation water and methods of controlling and managing soil salinity problems resulting from the practice of irrigated agriculture; (vii) Identify environmental issues related to the sector or any of the ensuing sub-projects requiring modification of ADB loan covenants to ensure subsequent appropriate resource management; (viii) Work with PMO and the PIUs to prepare/update a cost effective environmental management and monitoring plan for the rehabilitation and upgrading of the works for each sub-project, consistent with EIA recommendations so as to ensure minimal environmental effects both during and following the construction period; and (ix) Prepare and execute required and appropriate actions to mitigate any negative environmental impacts associated with construction activities in collaboration with PMO and the relevant PIU and all concerned stakeholders.

Water Management Specialist – International
Duration: 20 months.
Qualifications: The Water Management Specialist will have a Master's degree in irrigation engineering or water resources management with a minimum of fifteen years professional experience in managing both on-farm water management programs and irrigation service agencies and preparing and delivering meaningful support and training programs at all levels. At least five years of this experience should have been in South or Central Asia.
TOR: The Consultant will provide technical support to PMO for the Improved Water Resources Management component; the component consists of two sub components; (a) Basin-wide Irrigation and Drainage Management Institutional Optimization focusing on the Ferghana Valley, and (b) Improved On-Farm Water Management and WUA support in subproject areas located in both the Ferghana Valley and the Zarafshan River Basin. (i) working with the three BISAs serving the Ferghana Valley and the consultant's team, review and evaluate the administrative and management structure of the BISA, Main Canal Authorities (MCAs), Irrigation System Authority (ISAs), Pumping Plant Authority and Water Users Associations and their integration. Assess the BISA's and related organizations management and operational relationship with MAWR and other national, provincial and local government authorities. (ii) assess the communication and other operational procedures with regard to current operating efficiency of water management throughout the BISA system and with relevant national, provincial and local government authorities. Make recommendation to improve the communications and other operational procedures for the BISAs, other agencies and all other stakeholders throughout the Ferghana Valley. (iii) assess in particular the current procedures, reporting, coordination, and capacity (human and technical) for measurement and monitoring of water use throughout all levels the system (including both inter-farm and on-farm) including consideration of the transparency of data to stakeholders. Make recommendations to strengthen monitoring and measuring capacity throughout the system. (vi) assess in particular the current statutory provisions, procedures, reporting, coordination, and capacity (human and technical) for assessing water allocation and determining demand within the BISAs, including all relevant stakeholders. This would include delivery of bulk water supply as well as subsequent delivery of water throughout the inter-farm and to the on-farm system. Make recommendation to improve these procedures and supporting management structure. (v) assess the current use of computers, communications and other technology to support the operations and management of the BISAs and related organizations. Make recommendations to improve the use of technology to improve the operations and management of the BISAs to increase operational efficiency of the Ferghana Valley systems. (vi) compile findings and recommendations and develop and implement training programs targeted to BISA and ISA staff in the Ferghana Valley; (vii) Assess capacity and identify further needed capacity development to implement recommendations formulated to improve water management capacity within the BISAs serving the Ferghana Valley;.(vii) assume responsibility for and provide leadership to the separately contracted service for WUA and on-farm water management support in the irrigated commands of all selected sub-projects and assess on a continuing basis the adequacy of training and support materials proposed for use, the effectiveness of delivery mechanisms and the responses of both the WUAs and individual farmers to the program; (viii) Based on the detailed assessment conducted frame recommendations for PMO (and the third party service provider) as to how the

effectiveness of the WUA support could be strengthened and/or how it could be tailored to more fully meet the expectation and needs of the WUAs and the individual farmers: (ix) Suggest field level water management packages, tailored to the specific needs noted in the various sub-project areas, for use at the farm level and recommend extension type training techniques (and materials) relating to improved field channel layout, land leveling and improved irrigation application methodology for the use of the irrigators and share this with both PMO and the service provider; (x) Ensure that under the WUA support activities, a program of field measurement of crop water needs is demonstrated in each sub-project command to encourage demand based irrigation scheduling ; (xi) Identify affordable drip and sprinkler irrigation systems and design irrigation water management packages complete with on-farm water storage facilities if required which are specifically targeted toward orchard and vineyard owners and operators and ensure the WUA support service provider include their installation in selected sub-projects on a demonstration basis; (xii) Work with the team Agronomist and the service provider in the development of comprehensive farm level leaching practice appropriate to the applicable sub-projects, including practical guidelines for depths of leaching application based on the quality of the irrigation water available at the time of leaching and the current soil salinity levels to serve as the cornerstone practice related to proper management of saline soils; and (xiii) Report on a regular basis to MAWR, PMO and ADB regarding the execution and adequateness of both the institutional studies and the WUA support efforts.

Water Resources Engineer/Deputy Team Leader (Zarafshan) – National
Duration: 36 months

and

Water Resources Engineer/Deputy Team Leader (Fergana) – National
Duration: 50 months
Qualifications: The Water Resource Engineers/Deputy Team Leaders will have a Master's Degree in Civil Engineering and a minimum of 15 years experience in planning, designing and constructing lift irrigation projects in Uzbekistan.
TOR: (i) Assist the water resources/team leader by taking the leadership role for all project activities within the geographical area of responsibility; and in carrying out all aspects of his TOR relative to that area; (ii) In absence of the team leader assume full responsibility, including leadership of the consultant's team; (iii) Manage relationships with PIU, PMO MAWR, the relevant BISA, the and ADB as well as with other stakeholders including farmers and WUAs within the assigned area of responsibility; (iii) Ensure timely delivery and quality control of all required outputs at the PIU level including data bases to support feasibility studies for rehabilitation and upgrading of the remaining sub-projects, and field survey data and other information required to carry out detailed design and prepare the contract documents for all sub-projects and as-built drawings of completed works (including Narpay Machine Canal and the Besh Arik Sub-projects) within their respective areas of responsibility; (v) Assist in the preparation of detailed, time bound work plans for the preparation, design and tendering of all civil works contract packages envisaged for the works under the core sub-project in their assigned areas[12] assigning various team members working out of the PIU to key tasks; (vi) Provide technical support and guidance in all aspects of pumping plant, inter-farm

[12] These include among others, supply and install contracts for rehabilitation and upgrading of the Narpay and Besh Arik pumping plants, civil works contracts for the Narpay and Besh Arik inter-farm canals and civil works contract for drain rehabilitation within the Besh Arik irrigated command.

canal and drainage system design and tendering; (vii) Coordinate and supervise the preparation of tender documents for rehabilitating and upgrading pumping plants, inter-farm canals and agricultural drainage systems; (viii) In close coordination with PMO/PMI, assist in supervising the tendering process and subsequent evaluation of bids and preparation of award recommendations for all contracts for rehabilitation and upgrading; (ix) Provide coordination and oversight to ensure that monitoring and evaluation, resettlement, environmental and water management support aspects of all sub-projects are adequately addressed at all stages of project implementation; and (x) Monitor the progress of all planning, design and construction work ensuring that deadlines relating to delivery dates are met.

Pumping Plant Specialist – National
Duration: 51 months
Qualifications: The Pumping Plant Specialist shall have a Masters Degree in Mechanical Engineering and at least ten years experience in the planning, design and construction of irrigation pumping plants in Uzbekistan.
TOR: Asset the International Pumping Plant Specialist in carrying out his terms of reference; (ii) During the screening of additional sub-projects, visit the sites of the subprojects for assessment of present condition of pumps, motors and associated mechanical and electrical systems and collect required data; (iii) Carry out preliminary assessment and prepare preliminary designs and cost estimates for the rehabilitation work for the purpose of preliminary economic analysis ensuring that the short listed projects will fully meet the agreed selection criteria; (iv) During the feasibility phase visit the site for detailed assessment of the condition of the existing pumps, motors, piping, valves, overhead crane, rising mains and all other equipment relevant to the pumping stations; (v) Identify all repairs and replacement works necessary for the rehabilitation of pumping stations: (vi) Prepare a feasibility level design of for each proposed pumping station based on the calculated peak water requirements for the command area, finalize pump parameters and motor ratings, prepare basic drawings such as system flow diagrams, technical description of the works, a preliminary bill of quantities, cost estimates etc. as required for completion of feasibility reports; (vii) Review the design level field information gathered and verify the pump parameters proposed at the feasibility stage, particularly the pumping head, and carryout a detailed hydraulic analysis of the pumping system including the rising mains; (viii) Prepare specifications, data sheets and tender documents for the pumps and motors in coordination with Electrical Engineer: (ix) Work with the Procurement Specialist and carry out the pre-qualification process for pump and motor supply and install contract[13] for the purpose of international competitive bidding (ICB) for pump station rehabilitation and upgrading; (x) Design pump suction and discharge piping and prepare layout for realignment of piping outside the pump house including a common discharge header if appropriate and the rising mains;[14] (xi) Carry out preliminary analysis of hydraulic surge effects on the pumping system and design a surge protection system adequate to protect the pumping equipment and frame the

[13] Present thinking is to award a single supply and install contract for each pump-house under international competitive bidding which will include supply, fabrication and installation of all pumps, motors, piping, valves, electrical system as well as the rising mains and associated civil works related to pump foundations and building refurbishment.
[14] For each pump station going to final design a comprehensive testing program to determine the thickness of the rising main conduits will be carried out during the winter closure to provide an adequate basis for specifying those reaches to be replaced.

specifications for the same; (xii) Design an appropriate pump-house ventilation system and prepare associated equipment specifications, data sheets, drawings etc; (xiii) Undertake site visits to firm up the repair/ replacement works to intake works, over head crane, pump house building as envisaged at the time of feasibility and finalize work descriptions, quantities, specifications, drawings etc. as required for inclusion in bid documents; (xiv) Prepare a comprehensive bid document for rehabilitation of the pumping system on a supply and install basis including required drawings, data sheets and specifications; (xv) Assist PMO in providing clarifications to bidder's queries and in the technical evaluation of bids received; (xvi) Prepare a comprehensive operation manual for the pumping system and provide familiarization with operational procedures to all appropriate BISA an ISA staff; and (xvii) Provide supervision during the installation and construction of the works and continuing guidance during commissioning whenever possible.

Electro-mechanical Engineer – National
Duration; 26 months
Qualifications: The Electro-mechanical Engineer will have a master's degree in Electrical Engineering and at least 20 years experience in planning, design and installation of power supply for irrigation pumping plants in Uzbekistan and be particularly well versed regarding control circuitry associated therewith.

TOR: (i) working under the technical guidance of the Electrical Engineer assist him in carrying out applicable parts of his TOR; (ii) For those projects under feasibility study visit the sites of the sub projects for assessment of the present condition of the electrical system and collect required data with the help of National consultants and carry out an assessment and prepare feasibility level designs and cost estimates; (ii) During the design stage of the sub-projects, including Narpay and Besh Arik among others, .visit the site for detailed assessment of the existing condition of the electrical system serving the motors driving the pumps; (iv) Prepare technical specifications and data sheets. for electric motors as a part of bid documents; (v) Design all electrical systems for the pumping stations including cable sizing, cable layout, equipment layout in both the switchgear room and control room, lighting and an effective earthing (grounding) system; (vi) Design the control, protection and automation systems as well as the control desk: (vii) Prepare specifications, data sheets, single line diagrams, control schematics, drawings, bills of quantity, and schedule of rates for all aspects of the electrical systems and cost estimates for the same; (viii) Provide clarifications to bidder's queries and assist PMO in technical evaluation of bids; (ix) Participate in supervision of installation and construction as appropriate; (x) Prepare electrical related aspects of the Operation Manuals for each pumping station; and (xi) support PMO during the commissioning period and during initial operations.

Irrigation/Drainage Planner – National
Duration: 17 months
Qualifications: The Irrigation and Drainage Planner will have a Master's degree in civil/irrigation engineering and a minimum of fifteen years professional experience in the planning of medium and large-scale irrigation and drainage systems in Uzbekistan.

TOR: The Irrigation/Drainage Planner will work under the technical guidance of the Irrigation and Drainage Engineer and assist in carrying out those items in his TOR relative to preparation of the feasibility studies (ii) Review feasibility studies earlier

prepared for the Narpay and Besh Arik Sub-projects with a view to adoption of best practice innovations on ensuing projects; (iii) Based on both current and future cropping patterns, determine the crop water requirements for each crop included and based on current and improved future conveyance efficiencies, determine the irrigation requirements at the pumping plant, inter-farm canal head and key junctures along its course, and at each on-farm canal outlet; (iv) Prepare feasibility level designs for rehabilitating and upgrading the inter - farm canal system including all branches and associated control structures and water measurement facilities as well as other required supporting infrastructure including but not limited to cross drainage, emergency escapes and bridges for the commands of each of the remaining sub-projects; (v) Adopt and recommend innovations into the canal designs that will improve irrigation conveyance and operational efficiency and delivery flexibility ensuring equitable distribution of surface water throughout the command; (v) Review and recommend any changes in supply arrangements from the pumping plant to the inter-farm canals and from the inter-farm canals to on-farm canals to improve either operational efficiency or equity of distribution: (vi) Ensure that adequate flow measurement sites are provided through out the system to facilitate both system management and monitoring of system performance; (vii) identify all rehabilitation/upgrading required on the agricultural drainage system serving the irrigated command and package and cost the drainage rehabilitation work (viii) Work with the local agriculturist in making adequate provision for the water requirements of higher value crops which may be introduced into the cropping pattern in the future; and (ix) Prepare feasibility level designs[15] and cost estimates appropriate to establishing the feasibility of the sub-project.

Agronomist – National

Duration: 12 months.

Qualifications: The Agronomist will have a Master's degree in agriculture or irrigation agronomy and at least 15 years professional experience in planning major irrigated agriculture developments in Uzbekistan.

TOR: (i) Identify the major constraints faced by the farming communities in the inter-farm canal commands of all newly selected sub-projects which affect agricultural productivity including availability and dependability of the irrigation water supply and high water table and soil salinity problems.[16] (ii) Review the farming systems in place in each of the command areas with particular emphasis on improved wheat and cotton production technology, fodder crops, annual horticultural crops (vegetables) and perennial horticultural crops including orchards and vineyards. Assist the irrigation specialists in a review of both current and future cropping patterns with respect to water requirements with an objective to develop optimal strategies for cropping systems considering both the production quotas remaining in place and improvements in water availability, dependability and distribution to result from the rehabilitation and upgrading of the pumping plant, inter-farm distribution system and improved agricultural drainage; (iii) Provide the irrigation engineers and the economist with present, future without and

[15] Where possible, standard designs for structures of a type that will be repetitively used in the sub-project and subsequent sub-projects should be prepared in adequate detail to allow their repeated use in both subsequent feasibility studies and for final design. Standardization, where appropriate, will save considerable time and possible duplication of effort.

[16] This would be inclusive of all representative crops in all proposed sub- project areas including fruits, vegetables, fodder crops, oilseeds and maize as well as wheat and cotton.

future with cropping patterns (fully indicative of intensity), yields and production for each of the selected sub- project areas; (iv) Work with the monitoring and evaluation specialists in developing agriculture related monitoring and evaluation parameters for each sub-project and identify research requirements, if any, to ensure full achievement of project objectives; and (v) Assist the economist in assessing the linkages between rural poverty and irrigated agriculture in each of the sub-project area.

Resettlement Specialist – National
Duration: 17 months
Qualifications: The Resettlement Specialist shall have a Master's Degree in the Social Sciences and at least fifteen years experience in planning and implementing resettlement programs associated with irrigation projects in Uzbekistan.
TOR: (i) Assess all potential; resettlement impacts from the interventions foreseen under all proposed sub-project within the umbrella of ADB's resettlement policy; (ii) Review the resettlement framework prepared under the PPTA and ensure its continuing consistency with ADB guidelines in view of the nature of the additional sub-projects proposed and prepare necessary amendments as required; (iii) In collaboration with PMO and the concerned PIU, prepare any resettlement plans required under all components of the ensuing sub-projects; (iv) Develop detailed implementation arrangements to carry out any required resettlement activities under all sub-projects and assess the capacity within MAWR/PMO/PIU and other relevant agencies with respect to resettlement and prepare detailed capacity development programs for resettlement activities to be carried out under the sub-projects; (v) Assist PMO and PIU staff in complying with *ADBs Guidelines for Involuntary Resettlement* for those cases where resettlement or temporary disruption of production cannot be avoided; (vi) In such cases a resettlement plan, in accordance with the existing or suitably amended resettlement framework will be prepared for implementation by PMO and the relevant PIU; (vii) Train selected PMO and PIU staff with a view to strengthening of their ability to adequately oversee resettlement activities under each sub-project; and (viii) Review Government of Uzbekistan policies and practice relevant to resettlement, make recommendations for strengthening these and if necessary provide draft guidelines for Government's consideration.

Monitoring and Evaluation – National
Duration: 50 months
Qualifications: The Monitoring and Evaluation Specialist will have a Master's Degree in Economics/Social Science and at least fifteen years of professional experience, ten of which has been in developing and implementing monitoring and evaluation strategies and plans for agriculture and irrigation development interventions in Uzbekistan.
TOR: Working under the technical guidance of the international monitoring and evaluation specialist, assist him/her in carrying out his terms of reference including; (i) Assist PMO in the development and establishment of an appropriate monitoring and evaluation strategy and plans for rehabilitation and upgrading of pumping plants, inter-farm canal systems and agricultural drainage for all sub-projects including Narpay Machine Canal and the Besh Arik Lift Irrigation Sub-projects; (ii) The monitoring and evaluation systems referred to in (i) above will include input, progress, output and impact indicators and be computerized for each Sub-project to which they apply; (iii) Plan, design and supervise the conducting of baseline, intermediate and end-of Project

socio-economic surveys in the Narpay, Besh Arik[17] and other selected sub-projects; (iv) Identify a set of social and economic indicators that will serve as the benchmark for assessing the impact on farm income in both head and tail reaches of the inter-farm canals and in saline and non-saline areas if applicable. The indicators identified should also allow assessment of the direct and indirect social impacts of the Projects on the different categories of beneficiaries and vulnerable groups residing in or adjacent to the impacted irrigated commands; (v) Develop appropriate analytical methodology for the socio-economic impact assessment; (vi) Conduct in depth workshops for MAWR/PMO/PIU staff and for those sub-contractors who will carry out the surveys in the field, covering the survey design, sampling criteria, questionnaire requirements, interview methodology and survey analysis; and (vii) Supervise the work of those conducting the field surveys and analyzing assessment results.

Gender Specialist – National
Duration: 13 months
Qualifications: The Gender Specialist will have at least a Bachelor's degree in the social sciences and knowledge of gender issues in rural areas of Uzbekistan and a minimum of five years of professional experience in developing and implementing gender-specific and agriculture support activities in Uzbekistan.
TOR: (i) Review all documents relevant to the project including those prepared by the PPTA consultants; (ii) Assist PMO and the PIUs in planning, implementing and evaluating any sub-project interventions with which gender issues are associated; (iii) Ensure appropriate targeting of women from among the beneficiaries thus ensuring that they are able to benefit from all sub-project activities including the water management support activities; (iv) Liaise with PMO, PIU and the WUAs to enhance the ability of rural women to participate in all sub-project training, the FP demonstration programs and ensure their access to information related to irrigation systems and irrigated agriculture; (v) Assist the water management specialist and the irrigation/drainage engineer in field assessments of project area needs ensuring that the needs of the women in the sub-project areas (especially those of female headed households who are WUA members) are properly identified, reflected and addressed in the design of sub-project interventions and activities; (vi) Collaborate with the monitoring and evaluation specialist to ensure the socio-economic survey design is such that the views of women within the sub-project commands are properly reflected in the survey data base and in the survey results; and (vii) Work with the resettlement specialist to develop procedures for monitoring of resettlement impacts, if any, on women in the various sub-project command areas.

Environmental Specialist – National
Duration: 17 months
Qualifications: The Environmental Specialist will have a Master's degree in Environmental Science with at least fifteen years experience in conducting environmental screening/assessment of major water resources projects in Uzbekistan and will be fully familiar with ADB's Environmental Guidelines.
TOR: Working under the technical guidance of the international environmental specialist assist in carrying out his terms of reference including: (i) Prepare initial environmental examinations (IEEs) for each of the Sub- projects selected subsequent

[17] Full use will be made of the baseline data generated under the PPTA during the feasibility studies for the Narpay and Besh Arik Sub-projects.

to Narpay and Besh Arik to comply with ADB's environmental assessment guidelines and where applicable the specific requirements for sector type loans described therein; (ii) In cases where required conduct full environmental impact assessments which are in full compliance with ADB guidelines; (iii) Propose any environmental mitigation required and prepare environmental management and monitoring plans therefore and provide full details regarding the estimated costs of the required mitigation; (iv) Assess the capacity of MAWR/PMO/PIUs for environmental assessment and implementation of environmental management plans, and quantify training requirements for capacity building; (v) Develop training materials for PMO/PIU staff to support environmental protection measures and to monitor and mitigate potential environmental impacts; (vi) Assess the environmental capacity of the Uzbekistan government agencies involved in vetting and approving the environmental assessments and develop environmental assessment and review procedures and prepare detailed arrangements, agreeable to all parties, which may be used throughout the sector program to facilitate review of documents related to the various sub-projects; (vii) Work with relevant team members in an effort to provide particular emphasis with regard to environmental management of irrigation water and methods of controlling and managing soil salinity problems resulting from the practice of irrigated agriculture; (viii) Identify environmental issues related to the sector or any of the ensuing sub-projects requiring modification of ADB loan covenants to ensure subsequent appropriate resource management; (ix) Work with PMO and the PIUs to prepare/update a cost effective environmental management and monitoring plan for the rehabilitation and upgrading of the works for each sub-project, consistent with EIA recommendations so as to ensure minimal environmental effects both during and following the construction period; and (x) Prepare and execute required and appropriate actions to mitigate any negative environmental impacts associated with construction activities in collaboration with PMO and the relevant PIU and all concerned stakeholders.

Irrigation/Drainage Design Engineer – National
Duration: 28 months
Qualifications: The Irrigation/Drainage Design Engineer will have a Master's degree in civil/irrigation engineering and a minimum of fifteen years professional experience in the design and of medium and large scale irrigation systems in Uzbekistan.
TOR: (i) Working under the technical guidance of the international irrigation and drainage engineer provide the principal input and leadership for design of all irrigation and drainage rehabilitation and upgrading work, initially for the Narpay Machine Canal and Besh Arik Lift Irrigation sub-projects, followed by the works in all ensuing sub-projects making up batches two and three; (ii) Organize and supervise all topographic, profile and cross-section surveys required[18] to provide necessary input data for detailed design of the inter-farm canals, agricultural drains and associated works; (iii) Undertake the design of rehabilitation and upgrading or replacement works including but not limited to (a) inter-farm canal re-sectioning, grading, lining repair and replacement and rerouting if required, (b) head regulators, (c) cross regulators and on-farm canal outlet replacement/rehabilitation, (d) upgrading and/or replacement of existing water control and bifurcation structures and provision of additional structures

[18]To facilitate the collection of data, conducting of engineering and social surveys, carrying out geo-technical investigations and provide inspection and testing services during construction supervision (among others), the consultants contracted to conduct the services under Package 1 will be provided "provisional sums" to sub-contract these support activities

as required, (e) cross drainage structures such as inverted siphons and overshoots, (f) required bridges and culverts, and (g) associated agricultural drainage works; (iv) In the design of all structures and features of a repetitive nature, standard designs, pre-approved by the Team Leader and PMO should be used as appropriate to minimize duplicity of design inputs; (v) In coordination with the procurement specialist prepare all inputs required to facilitate tender document preparation for the rehabilitation and upgrading of all designated irrigation and drainage works.

Procurement Specialist – National
Duration: 25 months
Qualifications: The Procurement Specialist shall have at least a BS degree in Civil or Mechanical Engineering with 15 years experience in the procurement of civil and mechanical works and contract management on Government projects in Uzbekistan and be familiar with procurement under ADB/World Bank financed projects using international competitive bidding procedures.
TOR: Working under the technical guidance of the international procurement and contract specialist assist him to: (i) Prepare detailed procurement plans and packages and determine realistic time bound schedules for procurement, including parallel and sequential steps for procurement of civil and mechanical works from initial steps to the delivery of the services under the contracts for rehabilitation and upgrading work associated with the Narpay Machine Canal and Besh Arik Lift Sub-projects as well as those for all subsequent sub-projects to be included in Batches 2 and 3; (ii) Develop, in consultation with MAWR, PMO and ADB, the pre-qualification criteria, prepare notices of pre-qualification and pre-qualification documents and conduct the pre-qualification of international contractors in accordance with both MAWR procedures and ADB guidelines for applicable contract packages under each of the sub-projects; (iii) Under the direction of the Team Leader and using input from various specialists on the team prepare the international tender documents for the rehabilitation and upgrading of pumping plants at both Narpay and Besh Arik as well as all similar packages of work for subsequent sub-projects in FIDIC format agreeable to MAWR, PMO and ADB;[19] (iv) Organize and supervise appropriate pre-qualification exercises for local civil works contractors for the various civil works packages including both irrigation and drainage rehabilitation to be tendered under all sub-projects using procedures agreeable to MAWR and in accordance with applicable ADB Guidelines; (v) Under the direction of the Team Leader and using input from various specialists on the team prepare the tender documents for national competitive bidding for the rehabilitation and upgrading of irrigation and drainage works at both Narpay and Besh Arik as well as all similar packages of work for subsequent sub-projects in FIDIC format agreeable to MAWR, PMO and ADB;[20] (vi) Invite the pre-qualified bidders to submit bids and advise the committee established for evaluation regarding bid opening and the technicalities and process of evaluation in accordance with ADB's guidelines and MAWR requirements pertaining thereto; (vii) Advise on preparation of the summary of evaluation, recommendation for award and contract negotiations with successful bidders; and (vii) In collaboration with the team leader and the Project Director

[19] The documents to be prepared under (iii) above will include (a) invitation to bid, (b) instructions to bidders, (c) form of bid, (d) form of contract, (e) special and general conditions of contract, (f) drawings and specifications, (g) bill of quantities, (h) schedule of completion, and (i) all necessary addenda;
[20] The documents to be prepared under (iii) above will include (a) invitation to bid, (b) instructions to bidders, (c) form of bid, (d) form of contract, (e) special and general conditions of contract, (f) drawings and specifications, (g) bill of quantities, (h) schedule of completion, and (i) all necessary addenda;

familiarize PMO and PIU staff with the procurement procedures to be followed under subsequent contracts, including ADB's *Guidelines for Procurement* and undertake other activities targeted to developing the procurement and contracting capacity of MAWR staff.

Construction Engineers – National(Two)
Duration: 74 months
Qualifications: The Construction Engineers shall have MS degrees in Civil Engineering and at least twenty years experience in construction supervision of large-scale water resources/irrigation works in Uzbekistan.
TOR: (ii) Under the guidance of the Team Leader, assume full responsibility for construction supervision and contract administration; (ii) Ensure that all necessary field layout work to supplement the construction drawings is in place on site for the contractors' guidance. This includes all required location and elevation (line and grade) stake-out to facilitate the contractors' work on the irrigation system; (iii) Ensure comprehensive construction supervision and contract administration of both the supply and install contracts for pumping plant rehabilitation and the civil works contracts for rehabilitation and upgrading of the inter-farm canals and associated works and for rehabilitation of agricultural drains; (iv) Ensure that the consulting team fully assumes the role of "the Engineer" by undertaking all tasks defined under the FIDIC conditions of contract; (v) Carry-out/oversee all activities related to supervising construction works ensuring compliance to detail provided in the construction drawings and strict adherence to technical specifications; (vi) Oversee quality control testing methodology put in place, confirming its adequacy and ensuring that its employment is satisfactorily carried out including but not limited to (a) moisture-density testing of all soil materials to be used in construction and subsequent in-place density testing of compacted earth fills, (b) grain size distribution of all sand and gravel materials to be used in concrete and gravel materials to be used as filter materials, (c) routine slump tests on concrete used in all structures, (d) collection and compression testing of concrete cylinders, and (e) water quality testing of all water to be used in mixing concrete[21]; (vii) work closely with the pumping plant specialists in regard to all specialized inspection and testing related to new equipment and mechanical and electrical work associated with pumping plant rehabilitation; (viii) Ensure that detailed and quantitative progress reporting, including quantity surveys and measurement of all completed works generated in accordance with accepted methodology and standard practice, adequate to support the contractors requests for progress payments are routinely prepared and certified for subsequent payment; and (ix) Upon completion of construction activities under each contract, ensure that a comprehensive construction completion report, inclusive of as-built drawings as appropriate is prepared as a prerequisite to final payment under the contract for which the work has been performed.

Water Management Institutional Specialist – National
Duration: 24 months.

[21] To facilitate the collection of data, conducting of engineering and social surveys, carrying out geo-technical investigations and provide inspection and testing services during construction supervision (among others), the consultants contracted to conduct the services under Package 1 will be provided "provisional sums" to sub-contract these support activities.

Qualifications: The Water Management Institutional Specialist will have a Master's degree in irrigation engineering or water resources management with a minimum of ten years professional experience in institutional aspects of irrigation service agencies.
TOR: The Consultant will provide assist the international specialist in addressing the Basin-wide Irrigation and Drainage Management Institutional Optimization sub-component focusing on the Ferghana Valley. Specifically he or she will: (i) work with the three BISAs serving the Ferghana Valley and the international specialist to review and evaluate the administrative and management structure of the BISA, Main Canal Authorities (MCAs), Irrigation System Authority (ISAs), Pumping Plant Authority and Water Users Associations and their integration. Assess the BISA's and related organizations management and operational relationship with MAWR and other national, provincial and local government authorities. (ii) assess the communication and other operational procedures with regard to current operating efficiency of water management throughout the BISA system and with relevant national, provincial and local government authorities. (iii) assess in particular the current procedures, reporting, coordination, and capacity (human and technical) for measurement and monitoring of water use throughout all levels the system (including both inter-farm and on-farm) including consideration of the transparency of data to stakeholders (vi) assess in particular the current statutory provisions, procedures, reporting, coordination, and capacity (human and technical) for assessing water allocation and determining demand within the BISAs, including all relevant stakeholders. Make recommendation to improve these procedures and supporting management structure. (v) assess the current use of computers, communications and other technology to support the operations and management of the BISAs and related organizations. Make recommendations to improve the use of technology to improve the operations and management of the BISAs to increase operational efficiency of the Ferghana Valley systems. (vi) assist in compiling findings and recommendations and in formulating training programs targeted to BISA and ISA staff in the Ferghana Valley; (vii) Assess capacity and identify further needed capacity development to implement recommendations formulated to improve water management capacity within the BISAs serving the Ferghana Valley.

REPORTING REQUIREMENTS

The assignment's reporting requirements are outlined below. All reports shall be provided in English and Russian languages. The number of copies to be provided: (i) two hard copies to the Client in each language required, plus soft copy in Acrobat or MS Word format; (ii) one hard copy to ADB, plus soft copy in Acrobat or MS Word format.

(a) Inception Report. The inception report is designed to give the Client confidence that the assignment will be carried out as planned and as agreed on in the contract, and it should bring to the Client's attention major problems that might affect the direction and progress of the work. Inception report shall contain results of current situation analysis on following components:
   (i) Rehabilitation of irrigation infrastructure;
   (ii) Improved water resources management;
   (iii) Project management.

(iv)
Inception report shall be submitted to EA within 2 months after mobilization.

(b) Monthly Report.

(c) Quarterly Progress Reports. These reports keep the Client and ADB regularly informed about the progress of the project implementation. Progress reports shall be delivered on a quarterly basis. Progress reports may include a bar chart showing details of progress and any changes in the project implementation schedule. Photographs are a quick and easy way of conveying the status of a project, and their use in progress reports is encouraged. The quarterly progress reports also serve to set out the work program for the following months. The reports shall also include budgetary part describing necessary project budgetary and expenditures information.

(d) Midterm Report. This report shall be provided by Consultant after 24 months from the commencement date. Midterm report is required to inform the Client and ADB of preliminary results, alternative solutions, and major decisions that need to be made.

(e) Final Report. The final report is due at the completion of the assignment. This report will contain a complete review of activities undertaken, major results obtained, problems encountered, progress towards achieving the specific objectives of the project and impact on the beneficiaries. A section on recommendations and lessons learned will be included. The Client, ADB and Consultant should first discuss the report while it is still in draft form. After discussion and recommendations, the Consultant shall complete final report and submit it to the Client and ADB.

At the end of project activities, Consultant will organize a workshop to present and discuss the project achievements and provide suggestions and recommendations for possible follow up activities and development interventions, to be submitted to the Client and ADB.

SERVICES, FACILITIES AND EQUIPMENT TO BE PROVIDED BY THE CLIENT

The Client will provide adequate office accommodation at PMO and PIUs.

The Client will provide PMO and PIUs with adequate office accommodation with basic utilities. Consultants will equip the office premises; procure all the necessary office accessories, including fax, computers with appropriate software, printers, copiers, furniture, air conditioners etc. After the project completion, property ownership will pass to the Client. Consultants will be responsible for maintenance of equipment.

The Client will provide all available data, public reports, technical documents, engineering notes, studies, maps, legal documents, local services, equipment and facilities as are necessary and agreed for the services. These will be provided free of charge, unless the Client has to make additional expenses (in which case any such expense will be reimbursed to the Consultant).
All costs associated with the obtaining of technical and ecological expertise shall be paid directly by the PMO.

APPENDIX B

CONSULTING SERVICES
Preparation of Subsequent Sub-Projects, Detailed Design and Construction Supervision of All Sub-Projects and Operational Support for PMO

	Person Month Input per Year					
Specialist	Year 1	Year 2	Year 3	Year 4	Year 5	**Total**
Water Resources Engineer/Team Leader	6.0	11.0	11.0	8.0	8.0	44.0
Pumping Plant Specialist	6.0	8.0	5.0	4.0	2.0	25.0
Electrical Engineer	2.0	2.0	2.0			6.0
Irrigation/Drainage Engineer	6.0	8.0	4.0	4.0	2.0	24.0
Procurement Specialist	3.0	3.0	3.0			9.0
Project Economist	3.0	3.0			3.0	9.0
Monitoring and Evaluation Specialist	2.0	1.0	1.0			4.0
Environmental Specialist	2.0	2.0	1.0			5.0
Water Management Specialist	3.0	8.0	7.0		2.0	20.0
Total International Input (p/m)						**146.0**
Deputy Team Leader (Zarafshan)	7.0	11.0	11.0	7.0		36.0
Deputy Team Leader (Fergana)	7.0	11.0	11.0	10.0	11.0	50.0
Pumping Plant Specialist	7.0	11.0	11.0	11.0	11.0	51.0
Electro/Mechanical Engineer	6.0	8.0	4.0	4.0	4.0	26.0
Irrigation/Drainage Planner	3.0	11.0	3.0			17.0
Agronomist	3.0	9.0				12.0
Resettlement Specialist	3.0	11.0	3.0			17.0
Monitoring and Evaluation Specialist	6.0	11.0	11.0	11.0	11.0	50.0
Gender Specialist	3.0	10.0				13.0
Environmentalist	3.0	9.0	5.0			17.0
Irrigation/Drainage/Design Engineer	3.0	11.0	11.0	3.0		28.0
Procurement Specialist	3.0	11.0	11.0			25.0
Construction Engineers (2)		8.0	22.0	22.0	22.0	74.0
Water Management Specialist	3.0	6.0	6.0	6.0	3.0	24.0
Total National Input (p/m)						**440.0**

APPENDIX C

ESTIMATE OF UNIT RATE FEE
AND
OUT OF POCKET EXPENSES

Electrical Engineer /Kim Jeong-Seog

1 Unit Rate Fees of US$9,000 per person month for 6 months input = US$ 54,000
2 Per Diem of US$17/day for time spent in Uzbekistan
3 International Airfares in Economy Class

APPENDIX D
STAFFING SCHEDULE

TECH - 7 Personnel Schedule

No.	Name of Expert / Position	Professional Expert Input	Total p/m input		
			Home	Field	Total
International					
1	Byeong Ho CHEONG Water Resources Engineer/Team Leader	Home			
		Field		44	44
2	C.Madhawa Delpachitra Pumping Plant Specialist	Home			
		Field		25	25
3	Jeong Seog KIM Electrical Engineer	Home			
		Field		6	6
4	Joong-Gul PARK Irrigation/Drainage Engineer	Home			
		Field		24	24
5	Fauzy Ammari Procurement Specialist	Home			
		Field		9	9
6	J.V.M.Sarma Project Economist	Home			
		Field		9	9
7	S.M.F. Marikar Monitoring and Evaluation Specialist	Home			
		Field		4	4
8	W K B Elkaduwa Environmental Specialist	Home			
		Field		5	5
9	U N S Wickramaratchi Water Management Specialist	Home			
		Field		20	20
		Sub Total			146
Free Home Supporting Staff					
1	Lee Sang-Ik Project Director				
2	Kim Sung Ryul Geotechnical Engineer				
3	Kim Chae Soo Road Safety & Traffic Management Specialist				
National					
1	Amin Nasirov Deputy Team Leader (Zarafshan)	Home	36		36
		Field			
2	Ilhomjon Isamiddinov Deputy Team Leader (Fergana)	Home	50		50
		Field			
3	Oleg Glovatskiy Pumping Plant Specialist	Home	51		51
		Field			
4	Shavkat Rakhimov Electro/Mechanical Engineer	Home	26		26
		Field			
5	Ashirbek Muratov Irrigation/Drainage Planner	Home	17		17
		Field			
6	Alijon Umarov Agronomist	Home	12		12
		Field			
7	Orunbek Abuev Resettlement Specialist	Home	17		17
		Field			
8	Malika Ikramova Monitoring and Evaluation Specialist	Home	50		50
		Field			
9	Yulia Shirokova Gender Specialist	Home	13		13
		Field			
10	Sobitjon Mamatov Environmentalist	Home	17		17
		Field			
11	Vladimir Nasonov Irrigation/Drainage/Design Engineer	Home	28		28
		Field			
12	Ravshan Turgunbaev Procurement Specialist	Home	25		25
		Field			
13	Zair Jurabekov Construction Engineer 1	Home	37		37
		Field			
14	Shukrullo Urmanov Construction Engineer 2	Home	37		37
		Field			
15	Ismail Begimov Water Management Specialist	Home	24		24
		Field			
		Sub Total			440
		Total			586